



IBMRFO: Improved binary manta ray foraging optimization with chaotic tent map and adaptive somersault factor for feature selection

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ABSTRACT

Feature selection refers to the process of identifying the most relevant and valuable features or characteristics in a dataset, and it is an important step that can improve the performance and accuracy of predictive models for machine learning and data mining. Feature selection is crucial in efficiently reducing the number of features, thereby enhancing classification accuracy. This, in turn, alleviates computational burdens and elevates the overall performance of machine learning algorithms. This paper extended the conventional manta ray foraging optimization (MRFO) algorithm to an improved binary MRFO (IBMRFO) algorithm. IBMRFO serves as a search strategy for designing a wrapper-based feature selection approach. To commence, we introduce a novel mechanism, termed the chaotic tent map (CTM), aimed at initializing the population to fortify both the algorithm's exploitation capabilities and population diversity. Second, an adaptive somersault foraging (ASF) factor is tuned to avoid premature convergence. Finally, a binary mechanism is introduced to adapt the algorithm for binary feature selection dilemmas. Experiments are performed on 30 renowned UC Irvine machine learning repository datasets. The results of the student's t-test show that the classification algorithm has statistical significance. The findings unequivocally exhibit the superior performance of the proposed IBMRFO compared to other algorithms under scrutiny. Notably, the efficiency of the enhanced factors is assessed through rigorous testing. For post-publications supporting this work, readers can refer to <https://github.com/ELF-CHEUNG/IBMRFO>.

1. Introduction

Feature selection refers to identifying the most relevant and valuable features of a dataset or characteristics. It is an important step that can improve the performance and accuracy of predictive models in machine learning and data mining. Owing to the abundant availability of vast datasets, the domains of active matter, molecular and materials science, as well as biomedicine, have witnessed a surge in the utilization of machine learning and data mining techniques (Alber et al., 2019; Cichos, Gustavsson, Mehlig, & Volpe, 2020). Many datasets with high-dimensional feature spaces are created as machine learning models which get more complex (Faris et al., 2020). However, some of the features are redundant and pointless, which reduce classification accuracy and use up unnecessary computing resources. Therefore, by selecting and using only the most important features, algorithms would be more efficient, generalize better to new data, avoid overfitting and make more accurate predictions. Moreover, feature selection aids in

providing a better explanation for predictive models and unveiling concealed relationships within the data.

Feature selection sub-methods include three main categories that are filter, wrapper and embedding methods (Li et al., 2020). Specifically, filter methods use a statistical metric to rank features according to the calculated scores for each feature (Barddal, Enembreck, Gomes, Bifet, & Pfahringer, 2019). However, they may consume more computational resources. The wrapper methods use a classifier to evaluate the selection subset obtained by the selection algorithm and use feedback from the classifier to guide feature selection. Therefore, the accuracy of the wrapper methods is superior to that of the filtering method (Aličković & Subasi, 2017). Moreover, embedding methods are a particular case of packing methods since feature selection is considered as part of the training phase of machine learning (Huda & Banka, 2019; Osanaiye et al., 2016).

Meanwhile, feature selection can essentially be viewed as a global group and optimization problem. Specifically, a subset of the original

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dataset represents the solution to the optimization problem, and the exhaustive and heuristic search methods can solve the problem. However, the computational cost of exhaustive search methods is usually unacceptable compared to heuristic search methods, especially for high-dimensional datasets. Therefore, meta-heuristic search methods may be a more reasonable approach for solving the feature selection (Agrawal, Abutarboush, Ganesh, & Mohamed, 2021). Thus, it is necessary to conduct feature selections with effective methods. For example, Hegazy, Makhoul, and El-Tawel (2020) make an effort to strengthen the basic SSA's structural foundation in order to increase solution correctness, reliability, and convergence speed.

For the global search capability of feature selection algorithms, evolutionary computation has recently been applied to feature selection problems. Several researchers have used various population intelligence methods, such as cuckoo search (CS) (Yang & Deb, 2009), genetic algorithm (GA) (Aličković & Subasi, 2017), particle swarm optimization (PSO) (Huda & Banka, 2019), whale optimization algorithm (WOA) (Mirjalili & Lewis, 2016), sparrow search algorithm (SSA) (Mirjalili et al., 2017), harris hawks optimization (HHO) (Hussain, Neggaz, Zhu, & Houssein, 2021; Too, Liang, & Chen, 2021) and variants of these algorithms, for dealing with the feature selection problems.

Among these kind of algorithms, the manta ray foraging optimization (MRFO) algorithm, which has been shown to be both effective and easy to implement (Zhao, Zhang, & Wang, 2020). However, despite its widespread application, the conventional MRFO algorithm may suffer from certain limitations, such as inadequate exploitation of problem spaces. Furthermore, according to the No-Free-Lunch (NFL) theory, no single algorithm can be completely successful in tackling all optimization issues (Joyce & Herrmann, 2018). In particular, the conventional MRFO algorithm was originally developed for continuous optimization problems and may not be well-suited for feature selection tasks with binary solution spaces. Therefore, these conditions motivate us to improve the conventional MRFO so that making it more appropriate for feature selection tasks.

In this paper, we encapsulate the principal contributions as follows:

- We propose a formulated optimization problem that aims to concurrently minimize the number of selected features while enhancing the accuracy of classification.
- We propose an IBMRFO to solve the formulated joint feature selection problem. IBMRFO introduces a novel evolutionary population initialization chaotic tent map (CTM) mechanism, an adaptive somersault foraging factor (ASF) and a binary strategy to improve the performance of conventional MRFO to make it more appropriate for feature selection issues.
- Experiments are carried out using the UC Irvine (UCI) machine learning repository (Kelly, Longjohn, & Nottingham, 2023) to assess the performance of the proposed IBMRFO for feature selection, and the findings are compared to those of competing methods. Furthermore, the effectiveness of the proposed enhanced factors is validated.

The subsequent sections of this manuscript are structured as follows. Section 2 reviews the related work. Section 3 formulates the joint feature selection problem and proposes the IBMRFO. Section 4 shows the experiment setups and results, and Section 5 summarizes findings and conclusions.

2. Related work

This section reviews prior researches on feature selection methods across various approaches. Among the various feature selection methods can be divided into two categories: those based on evolutionary algorithms (EAs) and based on non-evolutionary algorithms (NEAs).

NEAs are a class of feature selection methods that do not rely on genetic operations such as mutation and crossover, have gained popularity due to their simplicity and efficiency (Li et al., 2020). However,

these methods employ a heuristic or greedy approach to select the most valuable features based on a performance metric, without using a population of candidate solutions as in EAs, such as those described in Effrosynidis and Arampatzis (2021), Neggaz and Fizazi (2022), Song, Zhang, Gong, and Gao (2022). Therefore, NEAs have limitations such as their inadequate ability to capture non-linear relationships, sensitivity to outliers, and overfitting potential when the number of features is extensive, which can result in biased results and poor performance on new and unseen data. In other words, NEAs are more suitable for low-latitude data in feature selection.

EAs use a population of candidate solutions and iterative selection mechanisms to search for an optimal feature subset that maximizes a fitness function. For example, the swarm intelligence-based feature selection method for amphetamine-type stimulants proposed in Yusof, Muda, and Pratama (2021), grasshopper optimization algorithm (GOA) (Saremi, Mirjalili, & Lewis, 2017), particle swarm optimization (PSO) (fang Song, Zhang, wei Gong, & yan Sun, 2021; Huda & Banka, 2019), hybrid particle swarm optimization (Sowan, Eshtay, Dahal, Qattous, & Zhang, 2023), genetic algorithms (GA) (Xue, Zhu, Liang, & Słowiak, 2021) and binary differential evolutionary algorithm (Wang, Xue, Liang, & Zhang, 2023) etc.. Moreover, when a large number of features are available and their correlation is unclear, EAs can select relevant features to simplify the model, and improve the corresponding performance and interpretability (Xue, Zhang, Browne, & Yao, 2016). Therefore, compared with NEAs, one significant advantage is that EAs could handle large and complex feature spaces effectively. In other words, EAs is more suitable for high-latitude data in feature selection. However, these algorithms can be sensitive to initial conditions, impacting the final solution, and generated solutions may not be interpretable, making it difficult to understand feature-target variable relationships.

Recently, there is a growing trend in using improved versions of swarm intelligence and EAs as feature selection methods. The chaotic dragonfly algorithm (CDA) (Sayed, Tharwat, & Hassanien, 2019) has been used to select the most discriminating features, and this algorithm bases on chaotic maps and the conventional dragonfly algorithm (DA). Houssein, Ibrahim, Neggaz, Hassaballah, and Wazery (2021) have proposed different electrocardiogram (ECG) signal descriptors based on one-dimensional local binary patterns, wavelets, higher-order statistics, and morphological information for feature extraction. There are several other studies have been proposed heuristics inspired by nature, such as the supervised confidence-based classification method for multi-class problems (Almugren & Alshamlan, 2019; Alshamaa, Chehade, & Honeine, 2018) have reviewed and compared recent hybrid approaches that use bio-inspired evolutionary methods as a wrapper method, and Hegazy et al. (2020) have attempted to improve the structure of basic SSA to enhance solution accuracy, reliability, and convergence speed. Other influential works in feature selection include of Chattopadhyay, Dey, and Basak (2020), Nadimi-Shahraki, Zamani, and Mirjalili (2022), Raschka (2018). Such as, a library that implements a variety of core algorithms and utilities for machine learning and data mining called Mlxtend proposed by Raschka (2018), an enhanced whale optimization algorithm used for feature selection in Covid-19 (Nadimi-Shahraki et al., 2022), MRFO used for feature selection in emotion recognition from audio signals (Chattopadhyay et al., 2020).

Conventional MRFO has been shown to outperform other evolutionary algorithms in some cases (Gaonkar, Singh, Prashanth, & Kuppli, 2020), and widely used in various domains, such as global optimization problems, feature selection, engineering design and other fields (Abdullahi, Hassan, Abdullahi, Aliyu, & Kim, 2023). Specially, Tang et al. (2023) propose a hierarchical structure and weighted fitness-distance balance selection to enhance population diversity in MRFO, the hierarchical approach allows different groups of individuals to search in different regions, improving solution diversity, Zhu, Wang, Zhou, and Yan (2023) proposes H-MRFO, which enhances the learning and adaptive capabilities through variable spiral factors of MRFO, progressive

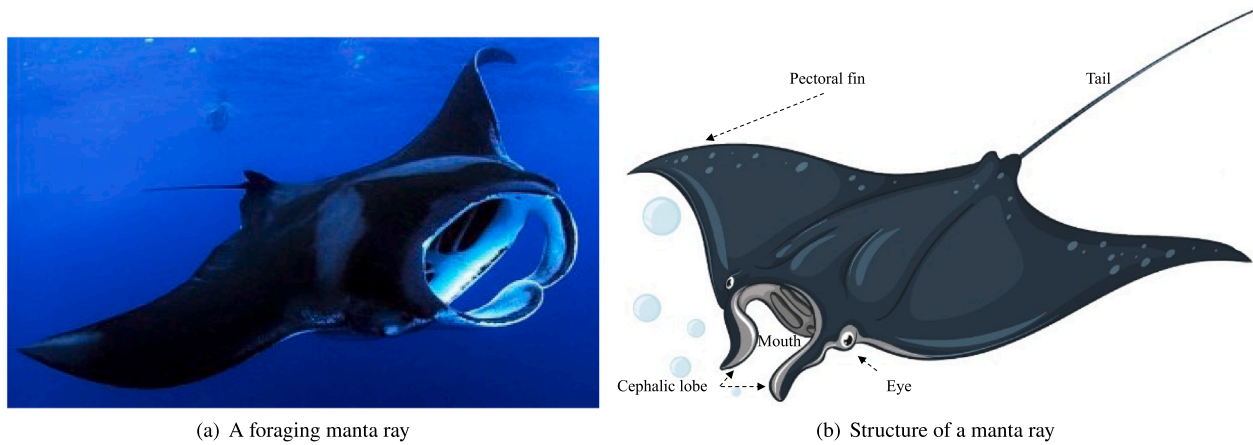


Fig. 1. A foraging manta ray and structure of a manta ray.

learning, and a matching game-based update mechanism. Additionally, Tang, Zhou, Han, and Xie (2021) propose a Modified MRFO in order to address the exploitation capability, tendency towards local optima, and decreasing population diversity. Moreover, conventional MRFO has advantages such as less parameters and easier convergence compared to other methods. Recently, a new approach based on conventional MRFO, developed following the food foraging nature of manta rays, has been proposed in Ghosh, Guha, Bera, Kumar, and Sarkar (2021), and previous researches have also focused on MRFO-based methods for feature selection. For example, Hassan, Abdullahi, Aliyu, Yusuf, and Abdulrahim (2022) use MRFO with a random forest classifier is used to detect network intrusions. Khodeir, Ababneh, and Alamoush (2022), Zhu, Wang, and Li (2022) have used MRFO in coverage optimization of wireless sensor networks. Baghdadi et al. (2022) have utilized MRFO with transfer learning framework to classifier breast cancer. In order to address the limitations of conventional MRFO and aim to improve its performance across different optimization problems and domains. Furthermore, MRFO has been applied in numerous domains such as image segmentation (Ma, Pereira, Oliva, Liu, & Kuo, 2023), mechanical design and Jusof et al. (2022), structural design (Zhong, Li, Meng, Li, & He, 2023) and even in screening COVID-19 (Dey et al., 2021; Houssein, Emam, & Ali, 2021). Other influential works include (Hemeida, Ibrahim, Mohamed, Alkhalaf, & El-Dine, 2021; Hu, Li, Wang, Wei, & Chang, 2022; Jain, Indora, & Atal, 2022; Liao, Zhao, & Wang, 2021). For instance, Hu et al. (2022) propose an enhanced MRFO to optimize complex curve shapes, benefiting computer-aided design applications, Jain et al. (2022) combines MRFO with a novel algorithm to create a model that detects glaucoma with high accuracy from retinal images, advancing early disease detection, Hemeida et al. (2021) applied the MRFO to power systems, the study optimizes renewable generator integration in distribution networks, improving energy efficiency and sustainability, and Liao et al. (2021) propose an improved MRFO for parameters Identification of magnetorheological dampers.

Although the methods mentioned above are capable of addressing feature selection problems in diverse applications, optimization algorithms may exhibit varying performance levels in different scenarios. Consequently, existing techniques may not effectively resolve all feature selection problems, which is supported by the NFL theorem (Joyce & Herrmann, 2018). Moreover, the performance of an optimization algorithm may vary depending on the feature selection application, such as feature selection in medical, Internet-of-Things and financial domains, etc. These domains include features with different data types (such as numerical or categorical), features with different levels of importance or relevance, features with varying levels of noise or redundancy, or features with complex relationships or interactions. Therefore, this motivates us to propose an IBMRFO that incorporates appropriate enhanced factors to address a broader range of feature selection challenges in this study.

3. Proposed IBMRFO for feature selection

3.1. Problem formulation

Within the context of this investigation, the core objective of feature selection is to achieve a dual outcome: reducing the selection of features while simultaneously enhancing classification accuracy. This pursuit encompasses a multi-objective optimization problem, where the intention is to strike a harmonious balance between these two contrasting goals (Abdollahzadeh & Gharehchopogh, 2022). A fitness function, which evaluates the quality of a given subset of features, is often used in conjunction with a metaheuristic algorithm to select features. As the algorithm iteratively explores different subsets of features, the fitness function assesses the quality of each subset. The algorithm can then use this information to search for the optimal feature subset. To address these two objectives, we define the fitness function by using the linear weighting approach, which is as follows:

$$f_{Fitness} = \epsilon \cdot E_r + \eta \cdot \frac{F_s}{F_a}, \quad (1)$$

where E_r denote the classification error rate of a specific classifier, which is equivalent to $(1 - accuracy)$. Additionally, we utilize F_s and F_a to represent the number of selected features and the total number of features, respectively. Meanwhile, we introduce ϵ and η as weight coefficients designed to harmonize and strike a delicate equilibrium between these twin objectives of maximizing classification performance and minimizing feature set size.

The k -nearest neighbour (KNN) methodology is employed in this study to build the wrapper method and measure classification accuracy. KNN is a well-known machine learning method that keeps all training data for classification purposes. More information about KNN can be found in Deng, Zhu, Cheng, Zong, and Zhang (2016). It is important to emphasize that in this work, the classifier itself does not constitute one of our main contributions. Rather, we leverage the outcomes of the classification process solely as indicators or indications for our research endeavors.

In the formulated feature selection problem, it is worth noting that the search space is characterized by nonlinearity and discreteness, potentially harboring numerous local minima. To tackle this challenge, we introduce the IBMRFO algorithm as a means to effectively address feature selection problems. In the subsequent sections, we present the intricate workings and finer nuances of this algorithm.

3.2. Conventional MRFO

Researchers have studied manta ray foraging behavior and have identified several key strategies that could be applied to feature selection algorithms. Fig. 1 illustrates the structure of a manta ray as well

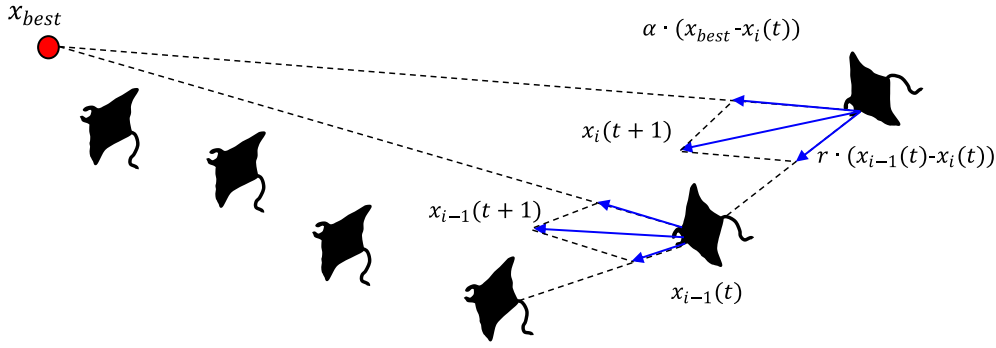


Fig. 2. Chain foraging.

as a foraging manta ray. Three different foraging mechanisms are used by manta rays: chain, cyclone, and somersault foraging.

In this algorithm, each manta ray serves as the distinctive representation of each candidate solution to the optimization problem. For the n -dimensional optimization problem, the i -th manta ray forages with a particular foraging method, and the position of the manta ray is denoted as x_i . Therefore, the position of a solution in the conventional MRFO algorithm can be written as: $x_i = (x_1, x_2, \dots, x_n)$. In each iteration, the position of each other manta ray which in the group is updated to generate a new solution. There are three update methods in conventional MRFO, which are detailed as follows.

3.2.1. Chain foraging

A two-dimensional (2D) space representation of chain foraging is provided in Fig. 2. In this behavior, all agents except the first one move towards the food source and the leading agent. Each agent updates its own optimal solution and incorporates the solution obtained by the preceding agent. The following is a representation of this chain foraging mathematical model:

$$x_i^d(t+1) = \begin{cases} x_i^d(t) + r \cdot (x_{best}^d(t) - x_i^d(t)) + A & i = 1, \\ x_i^d(t) + r \cdot (x_{i-1}^d(t) - x_i^d(t)) + A & i = 2, \dots, N, \end{cases} \quad (2a)$$

$$A = \alpha \cdot (x_{best}^d(t) - x_i^d(t)), \quad (2b)$$

$$\alpha = 2 * r * \sqrt{|\log(r)|}, \quad (3)$$

where r represents randomly generated numbers in the range $[0, 1]$, α is a weight coefficient, and $x_{best}^d(t)$ is the plankton with high concentration. $x_i^d(t)$ represents the position of the i -th individual at iteration t in the d -th dimension. In chain foraging, the best plankton location and the manta ray that came before it in the sequence determine the position update. However, when food is scarce, they may switch to more active foraging methods, such as cyclone foraging and somersault foraging.

3.2.2. Cyclone foraging

The process of cyclone foraging induces a spatial repositioning of each manta ray, guiding its movement not just towards the plankton source, but also along the trajectory of the leading manta ray in the group. The foraging behavior in a 2D space can be observed in Fig. 3(a). The update mechanism for cyclone foraging is mathematically described by Eqs. (4a) and (7a).

$$x_i^d(t+1) = \begin{cases} x_{best}^d + r \cdot (x_{best}^d(t) - x_i^d(t)) + B & i = 1, \\ x_{best}^d + r \cdot (x_{i-1}^d(t) - x_i^d(t)) + B & i = 2, \dots, N, \end{cases} \quad (4a)$$

$$B = \beta \cdot (x_{best}^d(t) - x_i^d(t)), \quad (4b)$$

$$\beta = 2e^{r_1 \frac{T-t+1}{T}} \cdot \sin(2\pi r_1), \quad (5)$$

where β is the weight coefficient, T represents the maximum number of iterations, and r_1 is a random number in the range $[0, 1]$. This foraging strategy enables manta rays to exploit the feasible region by focusing on the best solution found so far. Additionally, to encourage exploration, each manta ray is compelled to search for a new globally positioned solution by assigning a random reference position within the entire search space. This exploration mechanism is mathematically defined as follows:

$$x_{rand}^d = Lb^d + r \cdot (Ub^d - Lb^d), \quad (6)$$

$$x_i^d(t+1) = \begin{cases} x_{rand}^d + r \cdot (x_{best}^d(t) - x_i^d(t)) + C & i = 1, \\ x_{rand}^d + r \cdot (x_{i-1}^d(t) - x_i^d(t)) + C & i = 2, \dots, N, \end{cases} \quad (7a)$$

$$B = \beta \cdot (x_{rand}^d(t) - x_i^d(t)), \quad (7b)$$

where, the variable $x_{rand}^d(t)$ denotes a randomly selected position residing within the search space, respectfully confined by the upper bound u_i and the lower bound l_i . This placement is carried out without any specific bias or discrimination.

3.2.3. Somersault foraging

The somersault foraging behavior in conventional MRFO allows the algorithm to explore new regions in the solution space, both around the current position and symmetrical positions. This freedom of movement is guided by the best position obtained so far. As a result, they continuously update their positions around the best location. Fig. 3(b) provides a visual representation of this foraging behavior. The mathematical model can be defined by the following equation:

$$x_i^d(t+1) = x_i^d(t) + S \cdot (r_2 \cdot x_{best}^d - r_3 \cdot x_i^d(t)), i = 1, \dots, N, \quad (8)$$

where the parameter known as the somersault factor, denoted by S , governs the extent of somersaulting that a manta ray can achieve ($S = 2$). Within this process, both r_2 and r_3 are random values sampled uniformly from the interval $[0, 1]$. Consequently, the somersaulting behavior empowers the manta rays to explore novel regions, including symmetrical positions, guided by the previously attained optimal position.

Overall, the studies of manta ray foraging behavior has the potential to inspire new approaches to feature selection in machine learning and data mining. By incorporating critical strategies from manta ray foraging, such as adaptability and the ability to combine multiple techniques, researchers may be able to develop more efficient and effective feature selection algorithms.

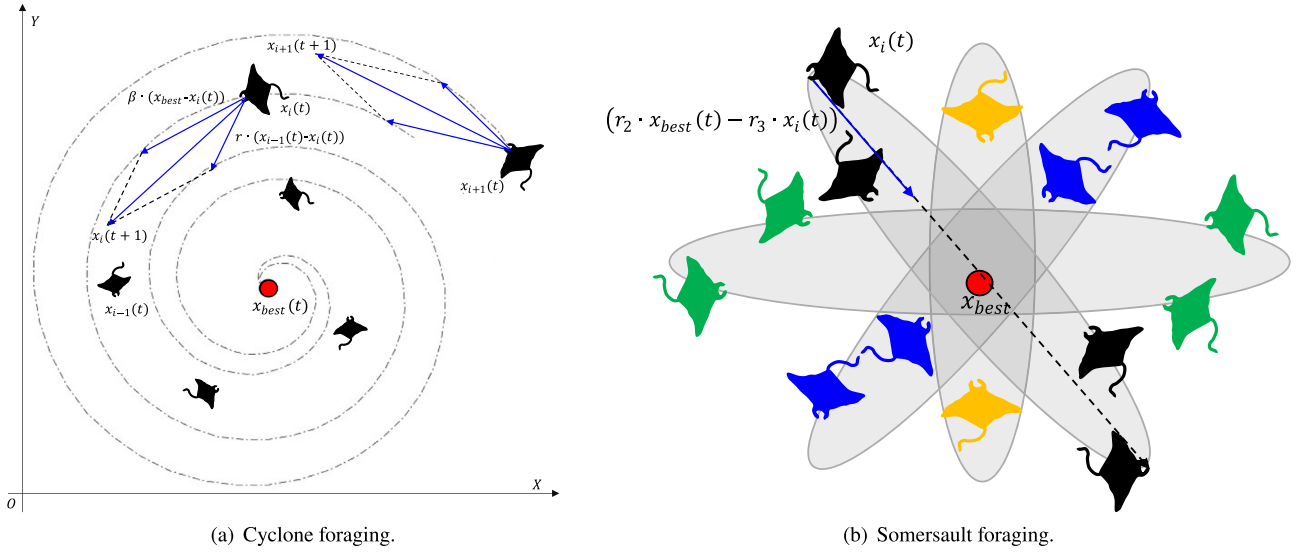


Fig. 3. Different foraging ways of manta rays.

3.3. The proposed IBMRFO

Conventional MRFO has been shown to have some drawbacks for some optimization issues. For optimization problems with large solution spaces, its exploration ability is relatively poor. Furthermore, it should be noted that the conventional MRFO algorithm, which was originally designed to tackle continuous problems, may not be applicable straightforwardly when confronted with feature selection problems. As a result, within the context of addressing the inherent intricacies encountered in formulaic feature selection, we put forth a novel approach known as the IBMRFO algorithm. This algorithm incorporates the crucial elements of the CTM and the ASF factor, thereby enhancing the overall performance of the conventional MRFO. Moreover, by employing a binary framework, the IBMRFO algorithm becomes better suited for effectively addressing the challenges posed by feature selection problems.

It is worth noting that the fundamental concept underlying the IBMRFO algorithm involves enabling the continuous search agents to traverse the search space dynamically and subsequently converting the obtained continuous solution into a binary representation. The algorithmic structure of the proposed IBMRFO is outlined in Algorithm 2, while the subsequent sections provide a comprehensive elucidation of the incorporated enhancement factors.

3.3.1. Chaotic Tent Map (CTM) mechanism for initialization population

In the conventional MRFO algorithm, the initial population information is typically established through the utilization of randomly generated data. However, this approach poses challenges in terms of maintaining population diversity, consequently hindering the algorithm's effectiveness in locating the optimal solution. Conversely, the CTM mechanism possesses inherent characteristics of randomness, ergodicity, and regularity. It serves the purpose of generating the initial population or acting as a perturbation throughout the algorithm's execution. By incorporating the CTM mechanism, the algorithm overcomes the limitations associated with being trapped in suboptimal local solutions, thus enhancing its search efficiency as compared to the original algorithm. Please refer to Algorithm 1 for a detailed description of the CTM mechanism.

Where a is the call threshold of the tent map and it is generally set as 0.5, plays a crucial role in the IBMRFO algorithm. In our approach, we incorporate a CTM as the initialization mechanism. To illustrate the effectiveness of this approach, we provide the Hillvalley in 32 selected datasets as an exemplary case of population initialization. The

Algorithm 1 Chaotic Tent Map (CTM) Mechanism.

Require: the size of population N_{pop} , solution dimension N_{dim} , chaotic tent map threshold a , low boundaries lb and up boundaries ub

Ensure: x

```

1: For  $i = 1$  to  $N_{pop}$ 
2:   For  $j = 1$  to  $N_{dim}$ 
3:     If  $rand < a$ 
4:        $x_{i,j} = \frac{rand}{a}$ 
5:     Else
6:        $x_{i,j} = \frac{(1-rand)}{a}$ 
7:    $x_i = lb + x_i \cdot (ub - lb)$ 
8: return mean of  $x$ 
9: For  $i = 1$  to  $N_{pop}$ 
10:  For  $j = 1$  to  $N_{dim}$ 
11:    If  $x_{i,j} < mean$ 
12:       $x_{i,j} = 0$ 
13:    Else
14:       $x_{i,j} = 1$ 
15: return  $x$ 

```

visualization in Fig. 4 showcases the population of 20 manta rays with a dimensionality of 100. It is worth highlighting that the red points represent the randomly initialized population, while the blue points correspond to the CTM-initialized population. A clear observation can be made: compared to the random initialization mechanism, the CTM mechanism exhibits superior distribution and randomness. Consequently, the implemented CTM initialization leads to a more even distribution of the population across the search space, ultimately contributing to enhanced optimization efficiency and solution accuracy within the algorithm.

3.3.2. Adaptive somersault factor (ASF) for foraging

Since the somersault coefficient is defined as a fixed value in the conventional MRFO algorithm, at the beginning of the algorithm, the manta ray can effectively expand the food search radius by somersaulting, and the algorithm can converge faster. However, as the algorithm continues to iterate, the fixed rotation radius may cause the manta ray to miss the best food location and fall into the local optimum. Therefore, to solve this problem, we propose ASF factor.

The somersault factor (S) is reformed using the best fitness function achieved thus far and the fitness function obtained by each agent

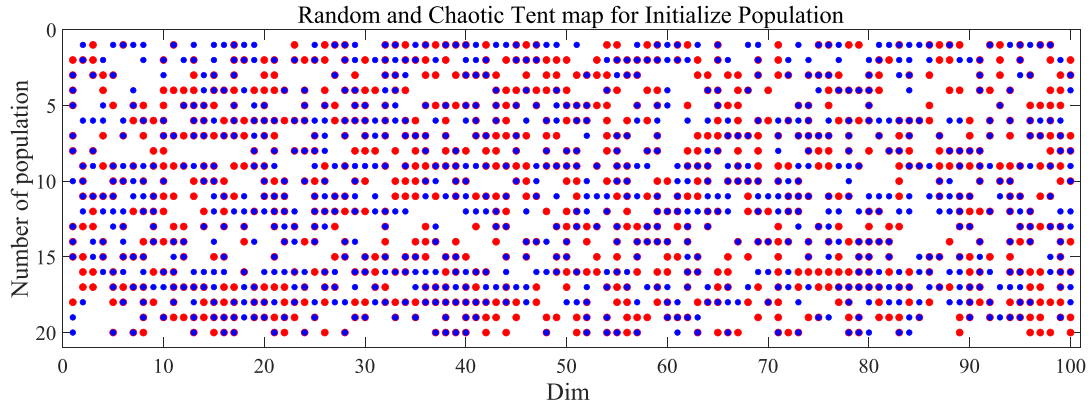


Fig. 4. Initialization of population: random and CTM mechanisms.

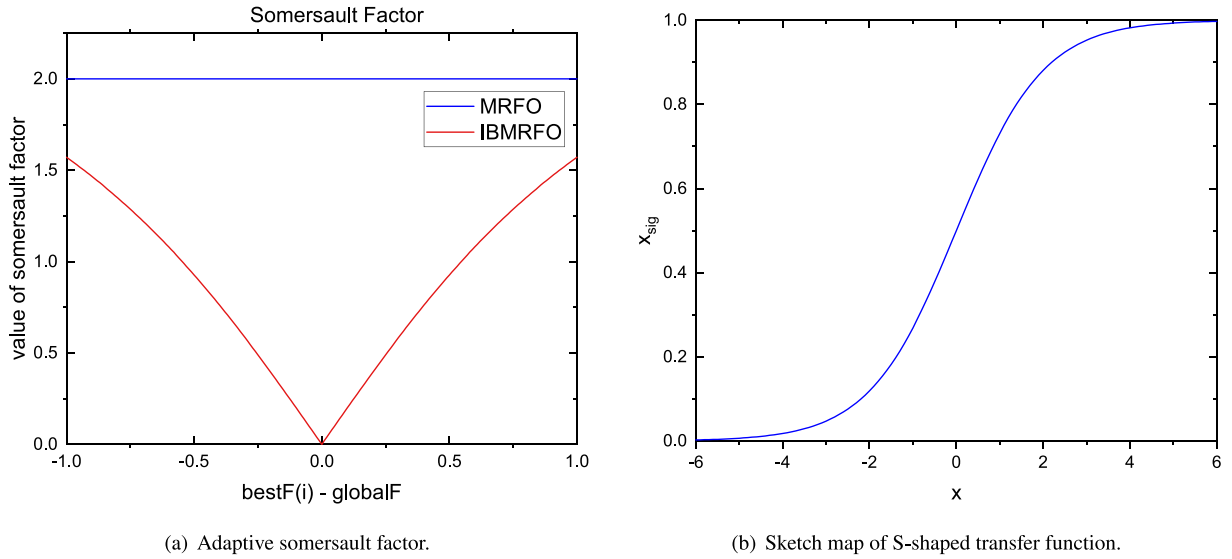


Fig. 5. Adaptive somersault factor and binary transfer function.

to avoid premature convergence. The somersault factor (S) has been adjusted to follow the inverse tangent function to maintain a smooth balance in sharing between considering prior terms from memory and the optimal agent placement reached thus far, as described as follows:

$$S = 2 \times \tan^{-1}(\text{abs}(\text{best}F(i) - \text{global}F)), \quad (9)$$

where $\text{best}F$ and $\text{global}F$ are the best functions of each agent and the global best one obtained so far, respectively. The $\tan^{-1}$ is the inverse tangent function. Fig. 5(a) shows the somersault foraging factor of conventional MRFO and IBMRFO, where the red line represents ASF factor and blue line represents the conventional somersault factor.

3.3.3. Binary mechanism sigmoid

The conventional MRFO algorithm updates the solutions in a continuous manner using Eqs. (2a), (4a), (7a), and (8) directly. However, when dealing with the feature selection problem, the solution space becomes discrete, which renders the conventional MRFO unsuitable for handling this scenario. To address this issue, a binary mechanism is introduced to map the continuous solutions to the discrete space, thereby making it applicable to feature selection problems. In this work, we employ the widely used S-shaped transfer function, known as the sigmoid function (Ghosh et al., 2021), to facilitate the mapping of solutions in both the conventional MRFO and IBMRFO algorithms. The details of this function are as follows. Additionally, the binary

mechanism is depicted as follows:

$$x_{sig} = \frac{1}{1 + e^{-x}}, \quad (10)$$

$$x_{binary} = \begin{cases} 1, & N_{random} \leq x_{sig}, \\ 0, & N_{random} > x_{sig}, \end{cases} \quad (11)$$

where x_{binary} is the converted binary solution of the feature selection problem, and N_{random} represents a randomly generated threshold. The binary mechanism employed in this study is illustrated in Fig. 5(b).

3.4. Feature selection based on IBMRFO

To address the formulated problem of feature selection using the proposed IBMRFO approach, we can draw an analogy as the manta ray. Hence, the solutions can be expressed as follows:

$$x = (M_1, M_2, M_3, \dots, M_{N_{dim}}), \quad (12)$$

where N_{dim} depicts the quantity of features while N_{pop} represents the number of individuals, correspondingly, the population of IBMRFO is expressed as follows:

$$pop = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{N_{pop}} \end{bmatrix}$$

Algorithm 2 Improved Binary Manta Ray Foraging Algorithm.

Require: the size of population N_{pop} , solution dimension N_{dim} , the max number of iterations T_{max} , the fitness function etc.

Ensure: x_{best}

- 1: Initialize the manta rays x according to Algorithm 1;
- 2: **For** $i = 1$ to T_{max}
- 3: Calculate the fitness function value of each manta ray according to Eq. (1); and obtain the best solution found so far x_{best} ;
- 4: **For** $i = 1$ to N_{pop}
- 5: **If** $rand < 0.5$ //Cyclone foraging
- 6: **If** $t/T_{max} < rand$
- 7: **Then** Perform the exploratory behavior of cyclone foraging according to Eqs. (5), (6), and (7a);
- 8: **Else**
- 9: **Then** Perform the exploratory behavior of cyclone foraging according to Eqs. (4a), (5), and (6);
- 10: **End If**
- 11: **Else** //Chain foraging
- 12: Perform the exploratory behavior of chain foraging according to Eqs. (2a) and (3);
- 13: **End If**
- 14: Map the continuous searching agents into binary spaces by using Eqs. (10) and (11);
- 15: Calculate the fitness of each individual according to Eq. (1),
- 16: **If** $f(x_i(t+1)) < f(x_{best})$;
- 17: **Then** $x_{best} = x_i(t+1)$;
- 18: **For** $i = 1$ to N_{pop} // Somersault foraging
- 19: Perform the exploratory behavior of chain foraging according to Eqs. (8) and (9) ;
- 20: Map the continuous searching agents into binary spaces by using Eqs. (10) and (11);
- 21: Calculate the fitness of each individual according to Eq. (1), **If** $f(x_i(t+1)) < f(x_{best})$;
- 22: **Then** $x_{best} = x_i(t+1)$;
- 23: **End For**
- 24: **End For**
- 25: Return the best solution found so far x_{best} .

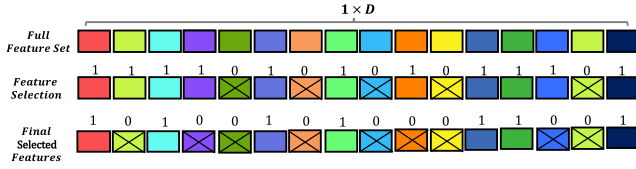


Fig. 6. Example of feature selection.

$$= \begin{bmatrix} M_1^1 & M_2^1 & M_3^1 & \dots & M_{N_{dim}}^1 \\ M_1^2 & M_2^2 & M_3^2 & \dots & M_{N_{dim}}^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ M_1^{N_{pop}} & M_2^{N_{pop}} & M_3^{N_{pop}} & \dots & M_{N_{dim}}^{N_{pop}} \end{bmatrix}, \quad (13)$$

The binary optimization problem in the feature selection task has only two possible solutions: 1 and 0. This work uses a $1 \times D$ vector to represent the solution; The dimension of the vector is determined by the number of elements in the original data set. Fig. 6 shows an example for feature selection, a selected binary element is displayed as 1; otherwise the corresponding attribute is linked to 0. To find the optimal feature combination, the developed binary version of the IBMRFO algorithm is adaptively executed in the feature selection process. One way to imagine the optimal feature combination is as an objective optimization problem with two requirements that must be met: the smaller the selected features, the better the classification performance (higher classification accuracy). In other words, the final quality of the solution is partly determined by the number of selected features and the classification accuracy.

3.5. Computational complexity

The complexity of the proposed IBMRFO is analyzed in this section. The time complexity of conventional MRFO depends on the number of variables, the number of individuals N_{pop} , and the maximum iteration

T_{max} . Somersault foraging, along with either cyclone foraging or chain foraging, is performed in each iteration. Therefore, the overall time complexity of IBMRFO consists of cyclone foraging, chain foraging, and somersault foraging. Thus, the overall time complexity of IBMRFO is given as follows:

This section is dedicated to analyzing the intricacies of the proposed IBMRFO. The time complexity of the conventional MRFO relies on various factors, including the number of variables, the number of individuals denoted as N_{pop} , and the maximum iteration denoted as T_{max} . In each iteration, a combination of somersault foraging, cyclone foraging, or chain foraging takes place. Consequently, the overall time complexity of the IBMRFO approach encompasses the time complexities of cyclone foraging, chain foraging, and somersault foraging. Hence, the comprehensive time complexity of the IBMRFO method is expressed as follows:

$$\begin{aligned} O(MRFO) &= O(T_{max}(O(cyclone + chain) + O(somersault))) \\ &= O(T_{max}(N_{pop} \cdot N_{dim} + N_{pop} \cdot N_{dim})) \\ &= O(T_{max} \cdot N_{pop} \cdot N_{dim}), \end{aligned} \quad (14)$$

Given the resemblance in structure between the proposed IBMRFO and the conventional MRFO, we can infer that the computational complexity of the IBMRFO approach is also determined to be $O(T_{max} \cdot N_{pop} \cdot N_{dim})$. As such, IBMRFO is not expected to require a significantly greater amount of computing time than conventional MRFO for solving a given feature selection problem, as both algorithms possess equivalent computational complexity.

4. Experiments and analysis

In this section, we embark on a comprehensive assessment of the performance of the proposed IBMRFO concerning its efficacy in addressing feature selection problems. Initially, we provide an introduction to the datasets and experimental setups employed in our evaluations. The test results from IBMRFO and a few comparison algorithms are then shown and discussed.

Table 1
Benchmark datasets.

No.	Dataset	Features	Instance	No.	Dataset	Features	Instance
1	Arrhythmia	278	452	16	Lung	72	326
2	Breastcancer	10	699	17	Lung-Cancer	56	32
3	BreastEW	30	569	18	Lymphography	18	148
4	Congress	16	435	19	M of N	14	1000
5	Dermatology	34	366	20	Movementlibras	90	360
6	Diabets	8	768	21	Semeion	265	1593
7	Exactly	13	1000	22	Sonar	60	208
8	Exactly-Two	13	1000	23	Spect	22	267
9	German	24	1000	24	Tic-tac-toe	9	958
10	Glass	9	214	25	Vehicule	18	846
11	Heart-StatLog	13	270	26	Vote	16	300
12	HeartEW	13	270	27	Vowel	10	901
13	Hepatitis	19	142	28	WDBC	31	569
14	Hillvalley	100	606	29	Wine	13	178
15	Ionosphere	34	351	30	Zoo	16	101

4.1. Datasets and setups

The datasets and experimental settings utilized are thoroughly described in this section.

4.1.1. Benchmark datasets

This study presents the benchmark datasets used in the analyzes and parameter configurations of several algorithms. We select 30 datasets from the UCI machine learning repository (Kelly et al., 2023). Table 1 displays the key details of these datasets.

4.1.2. Experiment setups

In the feature selection tests, the BCS (Rodrigues et al., 2013), BPSO (Dara & Banka, 2014), WOA (Mafarja & Mirjalili, 2018), BDE (Wang et al., 2023), H-PSO (Sowan et al., 2023), GF-NSGAI (Wang, Xue, Zhang, & Liang, 2021), BSSA (Shekhawat, Sharma, Kumar, Nayyar, & Qureshi, 2021), ESAMFO (Xu et al., 2021), SFLA (Liu et al., 2022), QBSSA (Xia et al., 2022), EGOA (Xu et al., 2023), MRFO and MRFO-S1 (Ghosh et al., 2021) are introduced as comparison algorithms for feature selection. Note that IBMRFO and MRFO both use the exact binary mechanism. MRFO is a binary version of conventional MRFO. The parameters of the conventional MRFO algorithm only have one somersault foraging factor, therefore, the parameters of IBMRFO are referred to the conventional MRFO. Both of them do not require additional parameter tuning work compared to other algorithms.

Indeed, parameter selection is a critical aspect of any optimization algorithm and can significantly influence its performance. This is particularly true for IBMRFO algorithm, however the IBMRFO algorithm has no additional parameters except the CTM threshold a . Therefore, we need to tune the CTM threshold of IBMRFO. However, since 30 datasets need to be adjusted in our work, this will require a lot of work. Since the BreastEW and HeartEW datasets are medium in size compared to other datasets, we selected them to adjust the important chaotic threshold a parameters of the proposed IBMRFO. The range of a is 0.1 to 0.9 with increments of 0.1, resulting in 18 different results. In addition, each value of a is run independently 30 times to avoid random bias, and the average results are presented. The results of the parameter optimization are displayed in Table 2. According to the results, IBMRFO achieves the best optimization results when $a = 0.5$. Therefore, we use $a = 0.5$ for all datasets.

Table 3 displays the critical parameter choices for these algorithms. Specific parameters are designated as the values that consistently perform well in the literature for the corresponding algorithms to compare features effectively.

Furthermore, since both the proposed IBMRFO algorithm and the above-mentioned comparative algorithms fall under the umbrella of meta-heuristics, the population size and the number of iterations exert a direct influence on their performance. Therefore, the population size and the number of iterations of different algorithms must be

Table 2

Parameter tuning results of chaotic threshold a in IBMRFO.

a	Dataset	Accuracy	Attributes	Fitness	$F1_W$	T-value
0.1	BreastEW	0.8354	5.733	0.167334	0.83303	7.64
	HeartEW	0.9479	3.900	0.052846	0.94788	49.13
0.2	BreastEW	0.8355	5.833	0.167300	0.83321	7.25
	HeartEW	0.9479	3.333	0.052657	0.94788	58.54
0.3	BreastEW	0.8354	5.933	0.167488	0.83305	7.04
	HeartEW	0.9479	3.133	0.052648	0.94782	69.59
0.4	BreastEW	0.8353	6.000	0.167687	0.83287	6.87
	HeartEW	0.9479	3.767	0.052801	0.94788	61.14
0.5	BreastEW	0.8357	5.733	0.167111	0.83330	7.59
	HeartEW	0.9480	3.633	0.052737	0.94790	59.73
0.6	BreastEW	0.8354	5.900	0.167536	0.83295	7.11
	HeartEW	0.9479	3.800	0.052812	0.94788	61.29
0.7	BreastEW	0.8356	5.900	0.167314	0.83321	7.05
	HeartEW	0.9480	3.400	0.052660	0.94790	59.93
0.8	BreastEW	0.8357	5.833	0.167114	0.83339	7.22
	HeartEW	0.9479	3.467	0.052720	0.94786	59.24
0.9	BreastEW	0.8351	5.800	0.167756	0.83259	7.53
	HeartEW	0.9479	3.933	0.052857	0.94788	56.81

Table 3

Key parameters of different algorithms.

No.	Algorithm	Parameters values
1	BCS	Discovery probability=0.25, $\alpha_1=1$
2	BPSO	$c_1 = 1.5$, $c_2 = 2.0$
3	BSSA	c_2 and c_3 random numbers over [0,1]
4	WOA	$\alpha_2 = 1$, $\beta_2 = 1$
5	BDE	$\delta = 0.1 * \text{Score}_{\max}$ scale factor $F = 0.5$ crossover rate $CR = 0.5$
6	H-PSO	crossover rate $c_5 = 1.5$, $c_6 = 2.0$
7	GF-NSGAI	div-the number of the divisions $div = 15$ the threshold $\theta = 0.6$
8	MRFO	$S = 2.0$
9	MRFO-S1	$S = 2.0$
10	ESAMFO	Cross-validation = 10
11	SFLA	$m = 5$, $\alpha = 6$, $\beta = 5$, $\delta = 2$
12	QBSSA	c_2 and c_3 random numbers over [0,1]
13	EGOA	$cMax = 1$, $cMin = 0.00004$
14	IBMRFO	adaptive S , threshold of CTM $a = 0.5$

consistent to ensure the fairness of the comparison. The population size and iteration count employed in this study were set to 20 and 150, respectively, for each algorithm considered. To mitigate any potential

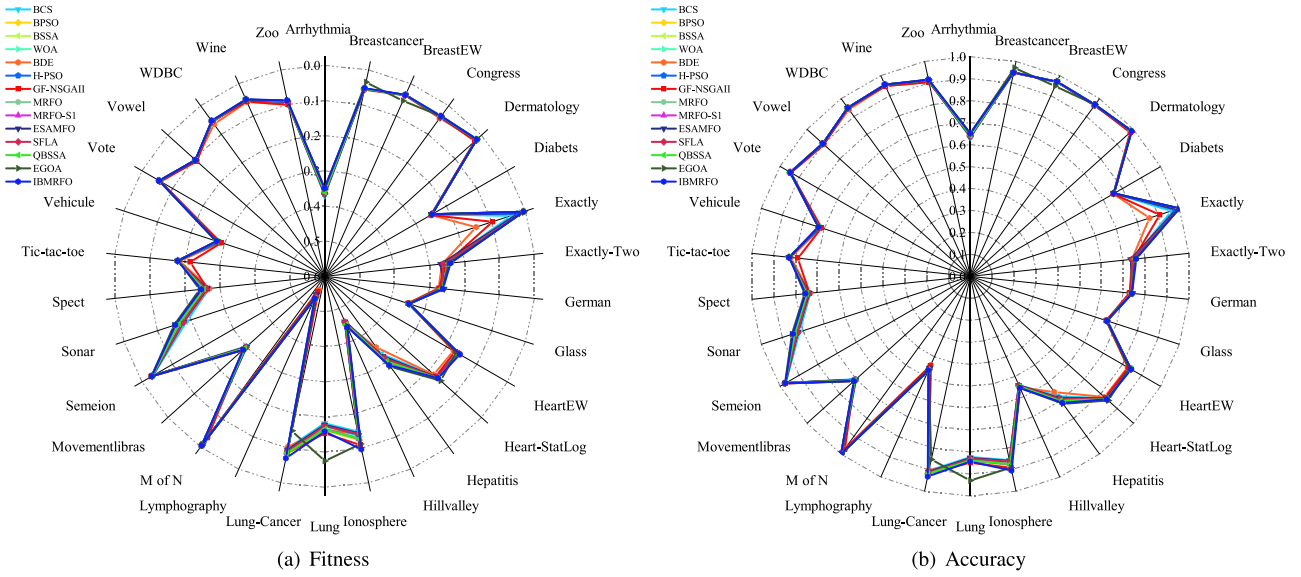


Fig. 7. The average fitness values and accuracy radar chart of different algorithms.

random biases, the central limit theorem guided us in independently executing each algorithm on the selected datasets a total of 30 times. The experimental setup entailed the use of an Intel(R) Core(R) I9-12900KF processor, accompanied by 64 GB of RAM. Python 3.9.12 was used as the programming language for implementation. Each dataset is randomly partitioned into a training set and a test set in the proportions of 70% and 30%, respectively. In the training phase, we use a simple learning algorithm, specifically K-Nearest Neighbors (KNN), and conduct 10-fold cross-validation to determine the classification error rate associated with the selected features in the training set. Additionally, the parameters in fitness function ϵ and η are set to 0.99 and 0.01, respectively.

4.2. Feature selection results

In this section, we provide an overview of the feature selection outcomes achieved by different algorithms, encompassing the evaluation of fitness function value, convergence rate, accuracy, number of selected features, F1 weighted score, and CPU time. The best results are bolded in Tables 4, 5, 6, 7 and 8. Additionally, to effectively elucidate the contrasting of fitness function value, convergence rate, accuracy, number of selected features, and F1 weighted scores among the different algorithms, we present the result through radar charts as shown in Fig. 7, where axes radiating from the center represent individual datasets.

4.2.1. Performance evaluation of different algorithms

Table 4 summarizes the numerical statistical results of the average fitness function value and standard deviation for each dataset, as obtained by different algorithms. Furthermore, we have chosen to specify the statistical test for each algorithm using the Student's t-test (Wang, Zhang, Liu, Lv, & Wang, 2014). A radar chart (Fig. 7(a)) was created to further illustrate the differences in fitness function values between the algorithms.

Based on the combined analysis of Table 4 and Fig. 7(a), it is obvious that the proposed IBMRFO algorithm provides the most favorable average fitness function values across 17 datasets. This might be the improvement factors we proposed come into play. In addition, among the 15 datasets, IBMRFO shows superior stability of the standard deviations of its fitness function values, highlighting its reliability in terms of feature selection. In addition, in the statistical test, Student's t-test was used to average the P-values of all algorithms are below the commonly

used significance level (e.g. 0.05), meaning that all algorithms are statistically significant. IBMRFO has the maximum average T-value, which means it is statistically better than other algorithms. Although its P-value is slightly higher than other statistically significant algorithms, IBMRFO is likely the superior algorithm given the significant differences in T-values.

It is important to highlight that the fitness function used in this study uses a linear weighting approach, effectively transforming the multi-objective optimization problem into a single-objective optimization problem. This approach optimizes both the accuracy of feature selection and the size of the selected features as a unified goal. Referring to the fitness function values obtained by various algorithms in Table 4 and the corresponding radar chart shown in Fig. 7(a). The edge of Fig. 7(a) corresponds to a fitness value of 0.0, with values closer to the edge indicating superior algorithmic fitness. The IBMRFO has demonstrated its superiority on most datasets and demonstrated its effectiveness and adaptability in addressing these issues by maintaining low fitness values on most datasets. In particular, the effectiveness of various amplification factors is further supported and discussed in Section 4.3.

To comprehensively evaluate the performance of the proposed algorithms, we present a series of convergence rate figures in this section. These figures offer insights into the optimization processes of various algorithms across different datasets. Notably, due to space constraints, these figures are presented in three parts. Each curve within these figures is extracted from the 15-*th* test iteration, providing a representative snapshot of algorithm behavior. In Figs. 8, 9, and 10, we show the convergence rates for each algorithms. These visualizations show the convergence speed of different algorithms on the various datasets. It should be emphasized that IBMRFO demonstrates superior performance across a majority of the datasets evaluated, not only exhibiting the most optimal convergence curve in 15 out of the 30 selected public datasets, but also achieving the best convergence curves in other datasets. The same optimal fitness function convergence value as other algorithms are obtained. This remarkable convergence ability clearly emphasizes the advantages of IBMRFO compared to the benchmark algorithms, especially in terms of average fitness values. This shows that our proposed IBMRFO algorithm has good performance, particularly in terms of average fitness values.

4.2.2. Features selection accuracy

The average classification accuracies of different algorithms are shown in Table 5. Table 5 indicates that IBMRFO achieves the best

Table 4
Average fitness function values obtained by different algorithms.

	BCS fitness $\pm$ std	BPSO fitness $\pm$ std	BSSA fitness $\pm$ std	WOA fitness $\pm$ std	BDE fitness $\pm$ std	H-PSO fitness $\pm$ std	GF-NSGAIH fitness $\pm$ std	MRFO fitness $\pm$ std	MRFO-SI fitness $\pm$ std	ESAMFO fitness $\pm$ std	SFLA fitness $\pm$ std	QBSA fitness $\pm$ std	EGOA fitness $\pm$ std	IBMRFO fitness $\pm$ std
Arrhythmia	0.3702 $\pm$ 0.0025	0.3562 $\pm$ 0.0025	0.3600 $\pm$ 0.0036	0.3639 $\pm$ 0.0031	0.3612 $\pm$ 0.0057	0.3656 $\pm$ 0.0045	0.3492 $\pm$ 0.0121	0.3612 $\pm$ 0.0028	0.3602 $\pm$ 0.0035	0.3670 $\pm$ 0.0039	0.3651 $\pm$ 0.0028	0.3601 $\pm$ 0.0043	0.3518 $\pm$ 0.0049	0.3488 $\pm$ 0.0077
Breastcancer	0.0565 $\pm$ 0.0004	0.0541 $\pm$ 0.0004	0.0549 $\pm$ 0.0004	0.0549 $\pm$ 0.0005	0.0579 $\pm$ 0.0035	0.0554 $\pm$ 0.0005	0.0535 $\pm$ 0.0011	0.0547 $\pm$ 0.0005	0.0546 $\pm$ 0.0005	0.0560 $\pm$ 0.0004	0.0554 $\pm$ 0.0005	0.0546 $\pm$ 0.0005	0.0535 $\pm$ 0.0006	0.0527 $\pm$ 0.0005
BreastEW	0.0352 $\pm$ 0.0001	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0364 $\pm$ 0.0008	0.0352 $\pm$ 0.0001	0.0362 $\pm$ 0.0008	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0	0.0352 $\pm$ 0.0
Congress	0.0426 $\pm$ 0.0023	0.0368 $\pm$ 0.0008	0.0377 $\pm$ 0.0014	0.0392 $\pm$ 0.0019	0.0439 $\pm$ 0.0044	0.0404 $\pm$ 0.0025	0.0394 $\pm$ 0.0032	0.0374 $\pm$ 0.0014	0.0376 $\pm$ 0.0014	0.0402 $\pm$ 0.0026	0.0392 $\pm$ 0.0025	0.0373 $\pm$ 0.0014	0.0363 $\pm$ 0.0006	0.0367 $\pm$ 0.0012
Dermatology	0.0200 $\pm$ 0.0015	0.0189 $\pm$ 0.0012	0.0178 $\pm$ 0.0010	0.0186 $\pm$ 0.0013	0.0204 $\pm$ 0.0028	0.0209 $\pm$ 0.0013	0.0250 $\pm$ 0.0079	0.0180 $\pm$ 0.0012	0.0184 $\pm$ 0.0011	0.0187 $\pm$ 0.0013	0.0185 $\pm$ 0.0012	0.0186 $\pm$ 0.0011	0.0170 $\pm$ 0.0012	0.0180 $\pm$ 0.0010
Diabets	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2514 $\pm$ 0.0031	0.2480 $\pm$ 0.0	0.2511 $\pm$ 0.0061	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0	0.2480 $\pm$ 0.0
Exactly	0.0426 $\pm$ 0.0267	0.0056 $\pm$ 0.0055	0.0068 $\pm$ 0.0083	0.0130 $\pm$ 0.0142	0.1468 $\pm$ 0.0788	0.0252 $\pm$ 0.0173	0.0977 $\pm$ 0.1323	0.0046 $\pm$ 0.0	0.0046 $\pm$ 0.0	0.0172 $\pm$ 0.0158	0.0076 $\pm$ 0.0092	0.0046 $\pm$ 0.0	0.0046 $\pm$ 0.0	0.0046 $\pm$ 0.0
Exactly-Two	0.2606 $\pm$ 0.0044	0.2411 $\pm$ 0.0049	0.2491 $\pm$ 0.0084	0.2510 $\pm$ 0.0083	0.2637 $\pm$ 0.0074	0.2546 $\pm$ 0.0078	0.2530 $\pm$ 0.0105	0.2468 $\pm$ 0.0086	0.2466 $\pm$ 0.0084	0.2685 $\pm$ 0.0027	0.2601 $\pm$ 0.0023	0.2436 $\pm$ 0.0070	0.2455 $\pm$ 0.0089	0.2394 $\pm$ 0.0
German	0.2718 $\pm$ 0.0026	0.2645 $\pm$ 0.0034	0.2655 $\pm$ 0.0035	0.2662 $\pm$ 0.0035	0.2745 $\pm$ 0.0066	0.2702 $\pm$ 0.0031	0.2723 $\pm$ 0.0066	0.2648 $\pm$ 0.0027	0.2646 $\pm$ 0.0038	0.2582 $\pm$ 0.0066	0.2671 $\pm$ 0.0032	0.2656 $\pm$ 0.0027	0.2635 $\pm$ 0.0033	0.2622 $\pm$ 0.0035
Glass	0.3472 $\pm$ 0.0003	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3510 $\pm$ 0.0037	0.3472 $\pm$ 0.0002	0.3505 $\pm$ 0.0049	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3471 $\pm$ 0.0	0.3477 $\pm$ 0.0016	0.3471 $\pm$ 0.0
HeartEW	0.1662 $\pm$ 0.0050	0.1562 $\pm$ 0.0009	0.1578 $\pm$ 0.0026	0.1589 $\pm$ 0.0037	0.1766 $\pm$ 0.0268	0.1638 $\pm$ 0.0049	0.1711 $\pm$ 0.0245	0.1564 $\pm$ 0.0013	0.1567 $\pm$ 0.0015	0.1595 $\pm$ 0.0035	0.1600 $\pm$ 0.0035	0.1565 $\pm$ 0.0014	0.1679 $\pm$ 0.0027	0.1560 $\pm$ 0.0008
Heart-StatLog	0.1734 $\pm$ 0.0039	0.1671 $\pm$ 0.0003	0.1675 $\pm$ 0.0014	0.1691 $\pm$ 0.0031	0.1825 $\pm$ 0.0090	0.1702 $\pm$ 0.0035	0.1748 $\pm$ 0.0068	0.1677 $\pm$ 0.0017	0.1674 $\pm$ 0.0013	0.1705 $\pm$ 0.0039	0.1698 $\pm$ 0.0038	0.1673 $\pm$ 0.0012	0.1561 $\pm$ 0.0010	0.1670 $\pm$ 0.0001
Hepatitis	0.3282 $\pm$ 0.0112	0.2943 $\pm$ 0.0097	0.3002 $\pm$ 0.0109	0.3083 $\pm$ 0.0093	0.3508 $\pm$ 0.0216	0.3154 $\pm$ 0.0101	0.3076 $\pm$ 0.0155	0.2997 $\pm$ 0.0098	0.2987 $\pm$ 0.0099	0.3184 $\pm$ 0.0120	0.3124 $\pm$ 0.0105	0.3004 $\pm$ 0.0081	0.2830 $\pm$ 0.0133	0.2875 $\pm$ 0.0141
Hillvalley	0.4599 $\pm$ 0.0030	0.4491 $\pm$ 0.0028	0.4505 $\pm$ 0.0033	0.4538 $\pm$ 0.0027	0.4497 $\pm$ 0.0049	0.4560 $\pm$ 0.0022	0.4428 $\pm$ 0.0098	0.4525 $\pm$ 0.0023	0.4519 $\pm$ 0.0039	0.4584 $\pm$ 0.0028	0.4575 $\pm$ 0.0025	0.4499 $\pm$ 0.0030	0.4448 $\pm$ 0.0033	0.4412 $\pm$ 0.0047
Ionosphere	0.1488 $\pm$ 0.0042	0.1231 $\pm$ 0.0064	0.1273 $\pm$ 0.0083	0.1341 $\pm$ 0.0059	0.1420 $\pm$ 0.0096	0.1395 $\pm$ 0.0057	0.1094 $\pm$ 0.0168	0.1251 $\pm$ 0.0075	0.1279 $\pm$ 0.0067	0.1445 $\pm$ 0.0044	0.1423 $\pm$ 0.0052	0.1275 $\pm$ 0.0070	0.1119 $\pm$ 0.0082	0.0969 $\pm$ 0.0095
Lung	0.1817 $\pm$ 0.0060	0.1622 $\pm$ 0.0060	0.1653 $\pm$ 0.0065	0.1720 $\pm$ 0.0076	0.1624 $\pm$ 0.0101	0.1756 $\pm$ 0.0049	0.1532 $\pm$ 0.0178	0.1651 $\pm$ 0.0074	0.1651 $\pm$ 0.0064	0.1776 $\pm$ 0.0050	0.1742 $\pm$ 0.0053	0.1648 $\pm$ 0.0070	0.0743 $\pm$ 0.0123	0.1587 $\pm$ 0.0083
Lung-Cancer	0.1016 $\pm$ 0.0092	0.0826 $\pm$ 0.0068	0.0856 $\pm$ 0.0082	0.0926 $\pm$ 0.0068	0.1050 $\pm$ 0.0144	0.0954 $\pm$ 0.0095	0.0862 $\pm$ 0.0166	0.0851 $\pm$ 0.0081	0.0875 $\pm$ 0.0083	0.0959 $\pm$ 0.0069	0.0919 $\pm$ 0.0078	0.0860 $\pm$ 0.0094	0.1506 $\pm$ 0.0087	0.0710 $\pm$ 0.0115
Lymphography	0.5489 $\pm$ 0.0086	0.5319 $\pm$ 0.0069	0.5301 $\pm$ 0.0068	0.5379 $\pm$ 0.0089	0.5580 $\pm$ 0.0140	0.5450 $\pm$ 0.0072	0.5523 $\pm$ 0.0139	0.5325 $\pm$ 0.0081	0.5333 $\pm$ 0.0072	0.5428 $\pm$ 0.0062	0.5373 $\pm$ 0.0092	0.5318 $\pm$ 0.0069	0.5328 $\pm$ 0.0089	0.5299 $\pm$ 0.0061
M of N	0.0134 $\pm$ 0.0099	0.0046 $\pm$ 0.0	0.0046 $\pm$ 0.0	0.0053 $\pm$ 0.0022	0.0341 $\pm$ 0.0258	0.0129 $\pm$ 0.0086	0.0323 $\pm$ 0.0496	0.0049 $\pm$ 0.0013	0.0049 $\pm$ 0.0013	0.0073 $\pm$ 0.0036	0.0065 $\pm$ 0.0032	0.0049 $\pm$ 0.0013	0.0046 $\pm$ 0.0	0.0049 $\pm$ 0.0013
Movementlibras	0.3036 $\pm$ 0.0029	0.2935 $\pm$ 0.0030	0.2947 $\pm$ 0.0035	0.2988 $\pm$ 0.0039	0.2957 $\pm$ 0.0054	0.3006 $\pm$ 0.0037	0.2867 $\pm$ 0.0085	0.2960 $\pm$ 0.0034	0.2951 $\pm$ 0.0037	0.3006 $\pm$ 0.0036	0.2986 $\pm$ 0.0033	0.2944 $\pm$ 0.0029	0.2879 $\pm$ 0.0050	0.2881 $\pm$ 0.0063
Semeion	0.0356 $\pm$ 0.0008	0.0323 $\pm$ 0.0009	0.0328 $\pm$ 0.0012	0.0344 $\pm$ 0.0008	0.0323 $\pm$ 0.0016	0.0344 $\pm$ 0.0010	0.0301 $\pm$ 0.0026	0.0333 $\pm$ 0.0008	0.0333 $\pm$ 0.0009	0.0344 $\pm$ 0.0009	0.0341 $\pm$ 0.0007	0.0331 $\pm$ 0.0006	0.0308 $\pm$ 0.0012	0.0317 $\pm$ 0.0016
Sonar	0.1883 $\pm$ 0.0056	0.1590 $\pm$ 0.0067	0.1649 $\pm$ 0.0092	0.1714 $\pm$ 0.0081	0.1739 $\pm$ 0.0137	0.1760 $\pm$ 0.0078	0.1611 $\pm$ 0.0172	0.1636 $\pm$ 0.0109	0.1640 $\pm$ 0.0080	0.1786 $\pm$ 0.0078	0.1769 $\pm$ 0.0082	0.1640 $\pm$ 0.0066	0.1490 $\pm$ 0.0091	0.1534 $\pm$ 0.0104
Spect	0.2698 $\pm$ 0.0073	0.2516 $\pm$ 0.0047	0.2536 $\pm$ 0.0058	0.2571 $\pm$ 0.0073	0.2701 $\pm$ 0.0120	0.2640 $\pm$ 0.0064	0.2650 $\pm$ 0.0129	0.2549 $\pm$ 0.0047	0.2550 $\pm$ 0.0059	0.2641 $\pm$ 0.0056	0.2620 $\pm$ 0.0052	0.2543 $\pm$ 0.0058	0.2426 $\pm$ 0.0053	0.2474 $\pm$ 0.0068
Tic-tac-toe	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1863 $\pm$ 0.0157	0.1795 $\pm$ 0.0	0.2156 $\pm$ 0.0275	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1795 $\pm$ 0.0	0.1827 $\pm$ 0.0123	0.1795 $\pm$ 0.0
Vehicle	0.2839 $\pm$ 0.0051	0.2783 $\pm$ 0.0028	0.2760 $\pm$ 0.0019	0.2790 $\pm$ 0.0040	0.2883 $\pm$ 0.0063	0.2856 $\pm$ 0.0038	0.2908 $\pm$ 0.0095	0.2771 $\pm$ 0.0030	0.2773 $\pm$ 0.0035	0.2794 $\pm$ 0.0040	0.2786 $\pm$ 0.0032	0.2776 $\pm$ 0.0030	0.2758 $\pm$ 0.0033	0.2767 $\pm$ 0.0031
Vote	0.0603 $\pm$ 0.0021	0.0558 $\pm$ 0.0007	0.0564 $\pm$ 0.0010	0.0568 $\pm$ 0.0012	0.0619 $\pm$ 0.0050	0.0589 $\pm$ 0.0026	0.0590 $\pm$ 0.0043	0.0563 $\pm$ 0.0008	0.0563 $\pm$ 0.0011	0.0582 $\pm$ 0.0019	0.0574 $\pm$ 0.0011	0.0560 $\pm$ 0.0008	0.0553 $\pm$ 0.0006	0.0553 $\pm$ 0.0005
Vowel	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1077 $\pm$ 0.0036	0.1054 $\pm$ 0.0	0.1116 $\pm$ 0.0062	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0	0.1054 $\pm$ 0.0
WDBC	0.0568 $\pm$ 0.0006	0.0543 $\pm$ 0.0004	0.0550 $\pm$ 0.0004	0.0553 $\pm$ 0.0005	0.0649 $\pm$ 0.0080	0.0558 $\pm$ 0.0006	0.0546 $\pm$ 0.0041	0.0546 $\pm$ 0.0005	0.0544 $\pm$ 0.0007	0.0561 $\pm$ 0.0004	0.0557 $\pm$ 0.0004	0.0546 $\pm$ 0.0006	0.0537 $\pm$ 0.0004	0.0528 $\pm$ 0.0004
Wine	0.0512 $\pm$ 0.0020	0.0495 $\pm$ 0.0012	0.0499 $\pm$ 0.0016	0.0500 $\pm$ 0.0015	0.0570 $\pm$ 0.0037	0.0514 $\pm$ 0.0020	0.0557 $\pm$ 0.0036	0.0490 $\pm$ 0.0009	0.0492 $\pm$ 0.0010	0.0500 $\pm$ 0.0013	0.0498 $\pm$ 0.0017	0.0492 $\pm$ 0.0010	0.0505 $\pm$ 0.0026	0.0489 $\pm$ 0.0006
Zoo	0.0961 $\pm$ 0.0034	0.0900 $\pm$ 0.0032	0.0904 $\pm$ 0.0033	0.0925 $\pm$ 0.0036	0.1017 $\pm$ 0.0095	0.0931 $\pm$ 0.0047	0.1003 $\pm$ 0.0101	0.0896 $\pm$ 0.0026	0.0894 $\pm$ 0.0029	0.0930 $\pm$ 0.0034	0.0920 $\pm$ 0.0034	0.0904 $\pm$ 0.0037	0.0883 $\pm$ 0.0020	0.0884 $\pm$ 0.0021
Mean	0.1816 $\pm$ 0.0044	0.1709 $\pm$ 0.0027	0.1723 $\pm$ 0.0034	0.1750 $\pm$ 0.0038	0.1869 $\pm$ 0.0113	0.1780 $\pm$ 0.0042	0.1796 $\pm$ 0.0154	0.1722 $\pm$ 0.0031	0.1723 $\pm$ 0.0031	0.1777 $\pm$ 0.0037	0.1762 $\pm$ 0.0033	0.1721 $\pm$ 0.0029	0.1684 $\pm$ 0.0040	0.1678 $\pm$ 0.0034
T-Value	5.9358	6.3704	6.4173	6.6453	5.7113	6.3353	8							

Table 5
Average classification accuracies achieved by different algorithms.

	BCS accuracy ± std	BPSO accuracy ± std	BSSA accuracy ± std	WOA accuracy ± std	BDE accuracy ± std	H-PSO accuracy ± std	GF-NSGAIL accuracy ± std	MRFO accuracy ± std	MRFO-S1 accuracy ± std	ESAMFO accuracy ± std	SFLA accuracy ± std	QBSSA accuracy ± std	EGOA accuracy ± std	IBMRFO accuracy ± std
Arrhythmia	0.6325 ± 0.0025	0.6451 ± 0.0025	0.6418 ± 0.0037	0.6379 ± 0.0030	0.6426 ± 0.0058	0.6361 ± 0.0045	0.6503 ± 0.0119	0.6407 ± 0.0027	0.6416 ± 0.0033	0.6357 ± 0.0039	0.6376 ± 0.0028	0.6415 ± 0.0040	0.6495 ± 0.0049	0.6508 ± 0.0072
Breastcancer	0.9476 ± 0.0005	0.9479 ± 0.0002	0.9478 ± 0.0003	0.9477 ± 0.0004	0.9464 ± 0.0034	0.9476 ± 0.0005	0.9478 ± 0.0004	0.9477 ± 0.0003	0.9478 ± 0.0003	0.9478 ± 0.0003	0.9479 ± 0.0003	0.9477 ± 0.0004	0.9479 ± 0.0002	0.9479 ± 0.0001
BreastEW	0.9704 ± 0.0003	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9694 ± 0.0008	0.9704 ± 0.0003	0.9685 ± 0.0018	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000	0.9705 ± 0.0000
Congress	0.9612 ± 0.0024	0.9663 ± 0.0010	0.9657 ± 0.0015	0.9641 ± 0.0021	0.9603 ± 0.0041	0.9627 ± 0.0026	0.9627 ± 0.0033	0.9657 ± 0.0016	0.9655 ± 0.0014	0.9636 ± 0.0029	0.9645 ± 0.0021	0.9655 ± 0.0018	0.9662 ± 0.0007	0.9656 ± 0.0014
Dermatology	0.9865 ± 0.0014	0.9868 ± 0.0013	0.9889 ± 0.0009	0.9879 ± 0.0015	0.9870 ± 0.0025	0.9854 ± 0.0014	0.9797 ± 0.0081	0.9883 ± 0.0012	0.9882 ± 0.0011	0.9878 ± 0.0013	0.9880 ± 0.0012	0.9876 ± 0.0011	0.9888 ± 0.0014	0.9882 ± 0.0011
Diabets	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7520 ± 0.0025	0.7545 ± 0.0000	0.7514 ± 0.0066	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000	0.7545 ± 0.0000
Exactly	0.9624 ± 0.0267	0.9990 ± 0.0054	0.9979 ± 0.0081	0.9917 ± 0.0140	0.8580 ± 0.0788	0.9797 ± 0.0171	0.9063 ± 0.1333	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9876 ± 0.0155	0.9970 ± 0.0091	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
Exactly-two	0.7412 ± 0.0037	0.7575 ± 0.0044	0.7501 ± 0.0079	0.7485 ± 0.0077	0.7392 ± 0.0061	0.7454 ± 0.0071	0.7465 ± 0.0095	0.7521 ± 0.0080	0.7523 ± 0.0079	0.7349 ± 0.0027	0.7412 ± 0.0015	0.7550 ± 0.0067	0.7536 ± 0.0079	0.7590 ± 0.0000
German	0.7315 ± 0.0027	0.7374 ± 0.0036	0.7377 ± 0.0034	0.7367 ± 0.0033	0.7297 ± 0.0063	0.7326 ± 0.0032	0.7288 ± 0.0067	0.7379 ± 0.0026	0.7381 ± 0.0037	0.7430 ± 0.0056	0.7363 ± 0.0032	0.7370 ± 0.0030	0.7382 ± 0.0034	0.7398 ± 0.0035
Glass	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6513 ± 0.0033	0.6538 ± 0.0000	0.6512 ± 0.0050	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6538 ± 0.0000	0.6535 ± 0.0009	0.6538 ± 0.0000
HeartEW	0.8371 ± 0.0046	0.8453 ± 0.0009	0.8441 ± 0.0023	0.8433 ± 0.0028	0.8271 ± 0.0269	0.8389 ± 0.0045	0.8306 ± 0.0244	0.8453 ± 0.0009	0.8450 ± 0.0011	0.8428 ± 0.0029	0.8425 ± 0.0026	0.8452 ± 0.0010	0.8348 ± 0.0033	0.8456 ± 0.0007
Heart-StatLog	0.8296 ± 0.0034	0.8358 ± 0.0006	0.8354 ± 0.0016	0.8334 ± 0.0032	0.8210 ± 0.0088	0.8324 ± 0.0035	0.8266 ± 0.0074	0.8351 ± 0.0018	0.8353 ± 0.0016	0.8321 ± 0.0041	0.8328 ± 0.0041	0.8355 ± 0.0014	0.8455 ± 0.0006	0.8359 ± 0.0004
Hepatitis	0.6729 ± 0.0113	0.7059 ± 0.0097	0.6998 ± 0.0108	0.6922 ± 0.0093	0.6512 ± 0.0213	0.6852 ± 0.0102	0.6914 ± 0.0156	0.7005 ± 0.0099	0.7016 ± 0.0097	0.6829 ± 0.0117	0.6888 ± 0.0103	0.7002 ± 0.0080	0.7171 ± 0.0132	0.7121 ± 0.0143
Hillvalley	0.5412 ± 0.0029	0.5507 ± 0.0027	0.5496 ± 0.0032	0.5463 ± 0.0026	0.5526 ± 0.0047	0.5444 ± 0.0022	0.5554 ± 0.0095	0.5475 ± 0.0022	0.5480 ± 0.0039	0.5427 ± 0.0028	0.5436 ± 0.0023	0.5501 ± 0.0031	0.5551 ± 0.0032	0.5566 ± 0.0044
Ionosphere	0.8547 ± 0.0040	0.8782 ± 0.0063	0.8746 ± 0.0080	0.8681 ± 0.0056	0.8620 ± 0.0092	0.8626 ± 0.0055	0.8914 ± 0.0162	0.8765 ± 0.0073	0.8736 ± 0.0065	0.8585 ± 0.0040	0.8608 ± 0.0051	0.8739 ± 0.0067	0.8892 ± 0.0079	0.9034 ± 0.0094
Lung	0.8227 ± 0.0058	0.8411 ± 0.0061	0.8380 ± 0.0067	0.8318 ± 0.0076	0.8430 ± 0.0103	0.8279 ± 0.0050	0.8483 ± 0.0176	0.8383 ± 0.0077	0.8383 ± 0.0063	0.8268 ± 0.0051	0.8303 ± 0.0054	0.8389 ± 0.0071	0.9290 ± 0.0124	0.8435 ± 0.0081
Lung-Cancer	0.9033 ± 0.0096	0.9207 ± 0.0069	0.9183 ± 0.0083	0.9113 ± 0.0068	0.9003 ± 0.0145	0.9087 ± 0.0097	0.9157 ± 0.0163	0.9187 ± 0.0082	0.9163 ± 0.0085	0.9090 ± 0.0071	0.9127 ± 0.0078	0.9177 ± 0.0094	0.8527 ± 0.0087	0.9313 ± 0.0117
Lymphography	0.4510 ± 0.0083	0.4678 ± 0.0072	0.4698 ± 0.0071	0.4617 ± 0.0090	0.4425 ± 0.0140	0.4544 ± 0.0075	0.4460 ± 0.0144	0.4673 ± 0.0084	0.4664 ± 0.0074	0.4570 ± 0.0064	0.4627 ± 0.0093	0.4680 ± 0.0073	0.4666 ± 0.0095	0.4696 ± 0.0066
M of N	0.9917 ± 0.0096	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9993 ± 0.0020	0.9714 ± 0.0254	0.9923 ± 0.0084	0.9726 ± 0.0501	0.9998 ± 0.0012	0.9998 ± 0.0012	0.9976 ± 0.0033	0.9983 ± 0.0028	0.9998 ± 0.0012	1.0000 ± 0.0000	0.9998 ± 0.0012
Movementlibras	0.6996 ± 0.0029	0.7081 ± 0.0031	0.7073 ± 0.0034	0.7035 ± 0.0040	0.7086 ± 0.0052	0.7016 ± 0.0038	0.7136 ± 0.0083	0.7063 ± 0.0033	0.7070 ± 0.0038	0.7024 ± 0.0035	0.7043 ± 0.0032	0.7072 ± 0.0030	0.7138 ± 0.0049	0.7126 ± 0.0058
Semeion	0.9705 ± 0.0007	0.9723 ± 0.0008	0.9721 ± 0.0013	0.9709 ± 0.0010	0.9751 ± 0.0014	0.9708 ± 0.0009	0.9730 ± 0.0024	0.9720 ± 0.0010	0.9721 ± 0.0008	0.9715 ± 0.0010	0.9718 ± 0.0008	0.9719 ± 0.0008	0.9739 ± 0.0012	0.9723 ± 0.0015
Sonar	0.8154 ± 0.0055	0.8438 ± 0.0066	0.8381 ± 0.0092	0.8319 ± 0.0080	0.8308 ± 0.0137	0.8269 ± 0.0080	0.8402 ± 0.0171	0.8395 ± 0.0109	0.8392 ± 0.0080	0.8255 ± 0.0077	0.8270 ± 0.0081	0.8392 ± 0.0066	0.8537 ± 0.0092	0.8484 ± 0.0107
Spect	0.7326 ± 0.0070	0.7497 ± 0.0045	0.7480 ± 0.0056	0.7447 ± 0.0071	0.7330 ± 0.0112	0.7379 ± 0.0063	0.7357 ± 0.0130	0.7470 ± 0.0045	0.7468 ± 0.0056	0.7381 ± 0.0055	0.7400 ± 0.0050	0.7474 ± 0.0057	0.7586 ± 0.0053	0.7539 ± 0.0066
Tic-tac-toe	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8216 ± 0.0167	0.8288 ± 0.0000	0.7899 ± 0.0296	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8288 ± 0.0000	0.8254 ± 0.0130	0.8288 ± 0.0000	0.8288 ± 0.0000
Vehicule	0.7194 ± 0.0051	0.7247 ± 0.0031	0.7272 ± 0.0020	0.7242 ± 0.0041	0.7156 ± 0.0061	0.7177 ± 0.0038	0.7113 ± 0.0100	0.7260 ± 0.0032	0.7257 ± 0.0034	0.7241 ± 0.0041	0.7248 ± 0.0031	0.7255 ± 0.0033	0.7267 ± 0.0033	0.7260 ± 0.0033
Vote	0.9439 ± 0.0019	0.9468 ± 0.0015	0.9473 ± 0.0013	0.9472 ± 0.0012	0.9433 ± 0.0044	0.9449 ± 0.0026	0.9430 ± 0.0039	0.9470 ± 0.0015	0.9474 ± 0.0014	0.9459 ± 0.0021	0.9473 ± 0.0011	0.9473 ± 0.0015	0.9474 ± 0.0010	0.9464 ± 0.0015
Vowel	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9003 ± 0.0036	0.9026 ± 0.0000	0.8955 ± 0.0071	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000	0.9026 ± 0.0000
WDBC	0.9474 ± 0.0007	0.9478 ± 0.0003	0.9477 ± 0.0003	0.9476 ± 0.0004	0.9393 ± 0.0080	0.9475 ± 0.0005	0.9466 ± 0.0036	0.9478 ± 0.0003	0.9477 ± 0.0003	0.9478 ± 0.0003	0.9478 ± 0.0004	0.9477 ± 0.0004	0.9479 ± 0.0001	0.9479 ± 0.0002
Wine	0.9537 ± 0.0023	0.9551 ± 0.0013	0.9549 ± 0.0014	0.9549 ± 0.0015	0.9490 ± 0.0029	0.9531 ± 0.0022	0.9479 ± 0.0040	0.9554 ± 0.0007	0.9554 ± 0.0007	0.9552 ± 0.0011	0.9548 ± 0.0015	0.9552 ± 0.0011	0.9540 ± 0.0024	0.9556 ± 0.0000
Zoo	0.9089 ± 0.0031	0.9141 ± 0.0032	0.9138 ± 0.0032	0.9120 ± 0.0035	0.9041 ± 0.0088	0.9114 ± 0.0046	0.9032 ± 0.0103	0.9146 ± 0.0028	0.9146 ± 0.0028	0.9117 ± 0.0032	0.9124 ± 0.0034	0.9137 ± 0.0034	0.9155 ± 0.0020	0.9155 ± 0.0020
Mean	0.8223 ± 0.0043	0.8319 ± 0.0028	0.8309 ± 0.0034	0.8283 ± 0.0037	0.8176 ± 0.0110	0.8253 ± 0.0042	0.8224 ± 0.0156	0.8309 ± 0.0031	0.8308 ± 0.0030	0.8261 ± 0.0036	0.8275 ± 0.0032	0.8310 ± 0.0029	0.8344 ± 0.0040	0.8346 ± 0.0034
Outperform	4(8)	7(7)	9(8)	5(6)	1(0)	4(6)	0(0)	6(7)	6(6)	6(6)	5(5)	6(7)	15(9)	16(12)

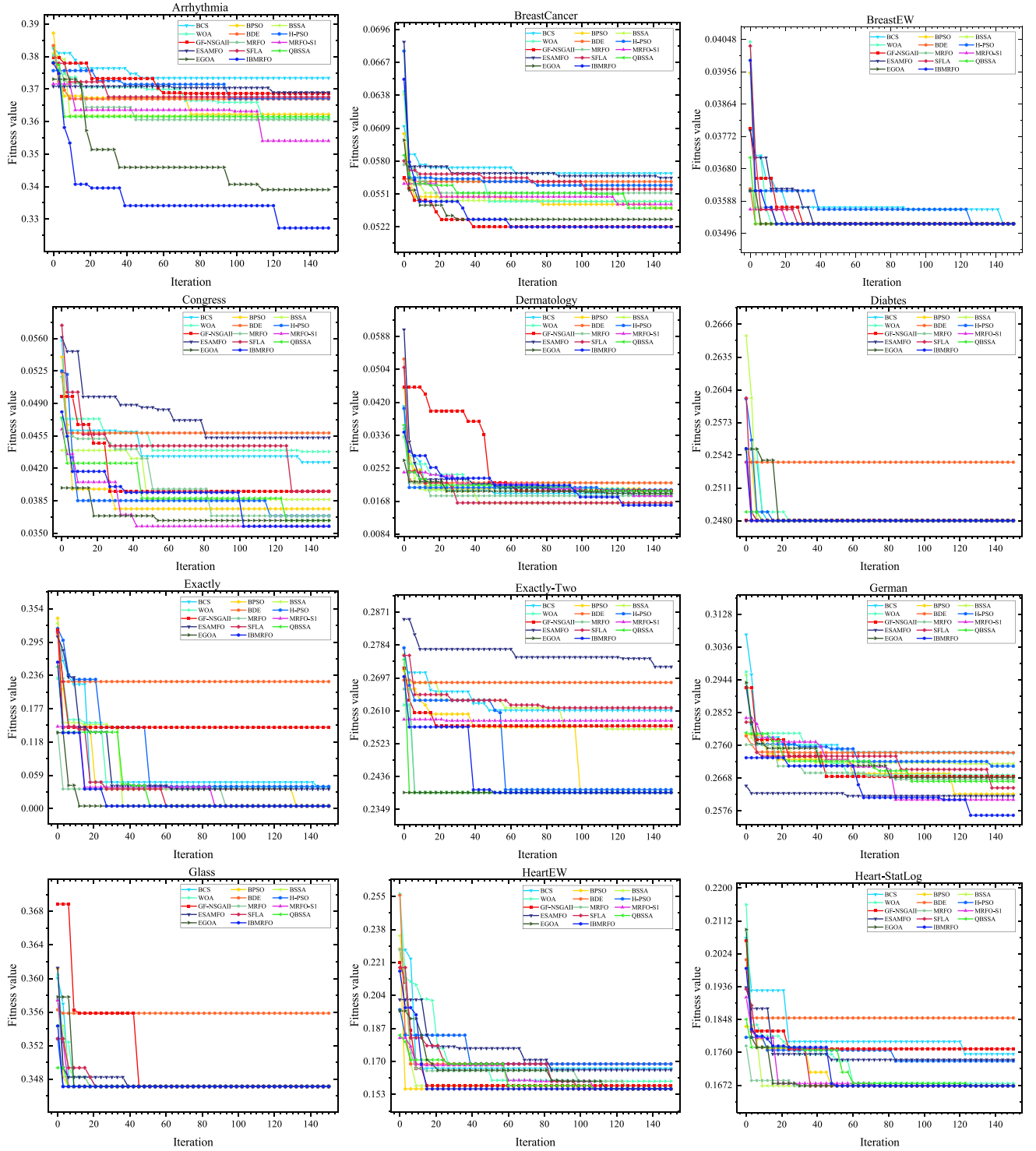


Fig. 8. Rate of convergence derived from diverse algorithms (Part 1).

average accuracy in 16 datasets and the lowest average accuracy standard deviation in 12 datasets. In other words, IBMRFO consistently outperforms others in terms of stability and effectiveness against a variety of other algorithms, highlighting its robustness and reliability in feature selection. The accuracy of each algorithm for 30 data sets is shown in Fig. 7(b), where the edge of Fig. 7(b) corresponds to an accuracy of 1.0, with values closer to the edge indicating superior algorithmic accuracy clues. The IBMRFO algorithm, characterized by its proximity to the 1.0 mark on most axes, consistently demonstrates high classification accuracy on most datasets. IBMRFO achieves the

highest accuracy, particularly for data sets such as BreastEW and Diabetes, which underlines its effectiveness. Furthermore, the minimal fluctuation of the IBMRFO line suggests commendable stability across different data sets.

4.2.3. Number of selected features

Table 6 shows a comparison of the average number of features selected by different algorithms in different datasets. Notably, GF-NSGAI outperforms other algorithms in this regard, achieving the lowest average number of selected features in 20 out of 30 datasets.

Table 6
Average number of selected features obtained by different algorithms.

	BCS attributes ± std	BPSO attributes ± std	BSSA attributes ± std	WOA attributes ± std	BDE attributes ± std	H-PSO attributes ± std	GF-NSGAIII attributes ± std	MRFO attributes ± std	MRFO-S1 attributes ± std	ESAMFO attributes ± std	SFLA attributes ± std	QBSSA attributes ± std	EGOA attributes ± std	IBMRFO attributes ± std
Arrhythmia	176.3 ± 11.3	136.6 ± 6.1	151.4 ± 20.4	150.7 ± 20.9	205.4 ± 9.2	148.1 ± 9.5	83.1 ± 27.6	152.9 ± 16.6	148.9 ± 20.7	175.7 ± 8.3	175.9 ± 6.4	144.6 ± 19.0	132.9 ± 7.1	85.9 ± 32.3
Breastcancer	14.0 ± 1.7	7.4 ± 1.3	9.4 ± 1.5	9.4 ± 1.6	14.3 ± 2.5	10.8 ± 1.4	5.3 ± 2.7	9.0 ± 1.8	8.6 ± 1.6	12.9 ± 1.1	11.4 ± 1.5	8.4 ± 1.7	5.9 ± 1.7	3.3 ± 1.3
BreastEW	6.0 ± 0.2	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.1 ± 0.4	6.0 ± 0.2	5.0 ± 1.1	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0
Congress	6.7 ± 1.4	5.5 ± 1.5	6.0 ± 1.6	5.8 ± 1.4	7.4 ± 1.4	5.6 ± 1.4	4.1 ± 1.2	5.5 ± 1.6	5.4 ± 1.4	6.6 ± 1.4	6.5 ± 1.5	5.0 ± 1.5	4.4 ± 1.6	4.3 ± 1.5
Dermatology	22.8 ± 2.1	19.5 ± 1.7	23.1 ± 1.9	22.7 ± 2.2	25.9 ± 1.9	21.8 ± 2.6	16.5 ± 2.6	21.8 ± 2.7	22.8 ± 2.5	22.7 ± 1.6	22.6 ± 1.4	21.4 ± 2.6	19.9 ± 2.3	21.4 ± 2.3
Diabets	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.8 ± 0.6	4.0 ± 0.0	4.0 ± 0.7	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0
Exactly	7.0 ± 0.4	6.0 ± 0.2	6.1 ± 0.3	6.3 ± 0.4	8.1 ± 1.0	6.6 ± 0.5	6.5 ± 0.7	6.0 ± 0.0	6.0 ± 0.0	6.4 ± 0.5	6.1 ± 0.3	6.0 ± 0.0	6.0 ± 0.0	6.0 ± 0.0
Exactly-two	5.7 ± 2.0	1.4 ± 0.8	2.2 ± 1.0	2.5 ± 1.5	7.3 ± 3.5	3.3 ± 1.4	2.7 ± 1.9	1.8 ± 1.0	1.7 ± 0.9	14.6 ± 1.9	5.1 ± 2.1	1.5 ± 0.7	2.0 ± 1.8	1.0 ± 0.0
German	14.2 ± 2.3	10.8 ± 1.5	14.0 ± 2.1	13.2 ± 2.1	16.6 ± 2.0	13.1 ± 2.0	9.2 ± 2.7	12.8 ± 2.3	12.6 ± 2.0	4.9 ± 2.2	14.4 ± 1.8	12.4 ± 2.6	10.2 ± 1.7	10.9 ± 2.3
Glass	4.1 ± 0.3	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	5.2 ± 1.0	4.0 ± 0.2	4.7 ± 0.9	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.0 ± 0.0	4.2 ± 0.6	4.0 ± 0.0
HeartEW	6.4 ± 1.4	4.0 ± 0.5	4.5 ± 1.0	4.9 ± 1.5	7.1 ± 1.0	5.6 ± 1.4	4.3 ± 1.2	4.1 ± 0.7	4.2 ± 0.9	5.0 ± 1.3	5.3 ± 1.4	4.2 ± 0.8	5.7 ± 0.9	4.0 ± 0.2
Heart-StatLog	6.1 ± 1.2	5.9 ± 0.4	5.8 ± 0.5	5.5 ± 0.8	6.8 ± 1.0	5.6 ± 0.9	4.1 ± 1.2	5.8 ± 0.6	5.7 ± 0.7	5.5 ± 0.7	5.6 ± 0.9	5.8 ± 0.5	4.1 ± 0.5	5.9 ± 0.4
Hepatitis	8.3 ± 1.8	6.0 ± 1.5	5.7 ± 1.7	6.8 ± 1.8	10.5 ± 1.7	7.2 ± 1.6	4.0 ± 1.3	6.1 ± 1.6	6.2 ± 1.4	8.5 ± 1.7	8.3 ± 1.7	6.7 ± 1.6	5.7 ± 1.3	4.7 ± 1.4
Hillvalley	57.5 ± 4.8	43.6 ± 5.3	45.9 ± 5.2	46.8 ± 7.6	66.9 ± 5.0	49.0 ± 5.0	27.2 ± 9.6	45.3 ± 7.5	44.4 ± 7.7	57.6 ± 5.0	57.4 ± 5.9	45.0 ± 7.1	43.6 ± 4.2	23.1 ± 9.5
Ionosphere	16.8 ± 2.7	8.8 ± 2.1	10.7 ± 2.4	11.6 ± 3.4	18.2 ± 2.5	11.8 ± 2.4	6.4 ± 3.1	9.6 ± 2.0	9.6 ± 2.2	15.2 ± 2.8	15.0 ± 2.3	9.0 ± 2.5	7.6 ± 2.2	4.4 ± 1.5
Lung	201.1 ± 14.6	156.9 ± 8.3	160.9 ± 15.7	179.8 ± 22.8	228.0 ± 10.5	170.2 ± 11.5	99.4 ± 31.5	164.4 ± 32.1	163.7 ± 24.2	201.2 ± 15.0	200.5 ± 11.3	172.9 ± 11.2	22.2 ± 3.1	123.0 ± 31.1
Lung-Cancer	33.0 ± 4.6	22.5 ± 3.6	26.5 ± 4.1	27.1 ± 5.5	35.5 ± 4.0	27.6 ± 3.9	14.9 ± 4.5	25.4 ± 4.6	25.9 ± 5.1	32.4 ± 3.0	30.4 ± 2.9	25.4 ± 5.7	154.4 ± 8.9	17.0 ± 6.8
Lymphography	9.7 ± 1.6	8.9 ± 1.1	9.4 ± 1.1	8.9 ± 1.7	11.0 ± 1.6	8.7 ± 1.4	6.9 ± 2.3	9.2 ± 1.1	9.2 ± 1.5	9.4 ± 1.4	9.7 ± 1.1	9.2 ± 1.3	8.5 ± 1.7	8.6 ± 1.5
M of N	6.8 ± 0.6	6.0 ± 0.0	6.0 ± 0.0	6.1 ± 0.3	7.6 ± 0.9	6.8 ± 0.5	6.6 ± 1.1	6.0 ± 0.2	6.0 ± 0.2	6.4 ± 0.5	6.3 ± 0.4	6.0 ± 0.2	6.0 ± 0.0	6.0 ± 0.2
Movementlibras	55.2 ± 3.9	40.6 ± 3.5	44.4 ± 7.2	46.6 ± 7.6	64.7 ± 4.6	46.6 ± 5.7	27.9 ± 7.1	47.2 ± 5.6	45.0 ± 6.9	53.7 ± 4.1	53.0 ± 4.2	40.6 ± 7.8	40.8 ± 4.4	31.5 ± 7.7
Semeion	169.1 ± 7.5	130.0 ± 8.2	137.7 ± 15.9	147.9 ± 17.3	203.6 ± 8.8	144.1 ± 10.8	88.9 ± 18.7	146.7 ± 12.6	149.0 ± 9.8	164.3 ± 8.8	163.7 ± 9.4	141.6 ± 16.1	130.2 ± 8.1	113.0 ± 14.4
Sonar	33.6 ± 4.6	25.6 ± 3.3	28.3 ± 4.3	29.7 ± 4.6	38.8 ± 3.7	27.8 ± 3.1	17.4 ± 4.8	28.4 ± 3.7	28.7 ± 5.4	34.9 ± 3.1	33.8 ± 4.1	29.0 ± 4.7	24.9 ± 2.4	20.1 ± 5.6
Spect	11.1 ± 1.7	8.5 ± 1.3	9.2 ± 1.6	9.7 ± 1.3	12.7 ± 2.2	9.9 ± 1.3	7.4 ± 2.2	9.6 ± 1.5	9.5 ± 1.7	10.6 ± 1.5	10.2 ± 1.2	9.4 ± 1.4	7.8 ± 1.0	8.3 ± 1.5
tic-tac-toe	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	8.7 ± 0.8	9.0 ± 0.0	6.8 ± 1.7	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	8.9 ± 0.5	9.0 ± 0.0
Vehicle	11.0 ± 1.2	10.4 ± 1.1	10.7 ± 1.2	10.7 ± 1.1	12.3 ± 1.1	11.0 ± 1.5	8.9 ± 1.6	10.5 ± 1.2	10.4 ± 1.2	11.3 ± 1.1	11.0 ± 0.9	10.5 ± 1.2	9.6 ± 0.9	9.8 ± 0.9
Vote	7.5 ± 1.5	5.0 ± 2.6	6.7 ± 2.1	7.3 ± 1.3	9.2 ± 1.7	7.1 ± 1.6	4.1 ± 2.5	6.1 ± 2.4	6.8 ± 1.9	7.4 ± 1.1	8.3 ± 0.8	6.1 ± 2.2	5.1 ± 2.2	3.5 ± 2.4
Vowel	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.6	9.0 ± 0.0	8.1 ± 0.9	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0	9.0 ± 0.0
WDBC	14.8 ± 2.0	8.1 ± 1.6	9.8 ± 1.3	10.8 ± 1.6	15.0 ± 2.4	12.0 ± 1.7	5.2 ± 2.8	9.0 ± 1.4	8.3 ± 2.1	13.6 ± 1.4	12.4 ± 1.5	9.0 ± 1.8	6.8 ± 1.3	3.8 ± 1.2
Wine	7.0 ± 1.0	6.5 ± 1.0	6.8 ± 0.9	7.0 ± 0.9	8.5 ± 1.5	6.4 ± 0.9	5.3 ± 0.9	6.3 ± 0.7	6.6 ± 0.9	7.3 ± 0.9	6.5 ± 0.9	6.2 ± 0.9	6.3 ± 0.6	6.3 ± 0.8
Zoo	9.6 ± 1.3	7.8 ± 0.9	8.1 ± 1.1	8.6 ± 1.0	10.7 ± 1.7	8.6 ± 1.2	7.1 ± 1.4	8.1 ± 0.9	7.8 ± 1.1	8.9 ± 1.2	8.4 ± 0.9	7.9 ± 1.2	7.4 ± 1.0	7.6 ± 1.2
Mean	31.4 ± 2.7	24.1 ± 2.0	26.0 ± 3.2	27.3 ± 3.7	36.2 ± 2.7	26.9 ± 2.5	16.7 ± 4.7	26.3 ± 3.5	26.2 ± 3.5	31.0 ± 2.39	30.7 ± 2.24	25.86 ± 3.20	23.67 ± 2.07	18.9 ± 4.3
Outperform	2(4)	5(16)	3(7)	2(6)	0(0)	1(4)	20(1)	4(9)	4(6)	2(5)	2(6)	4(6)	4(5)	11(13)

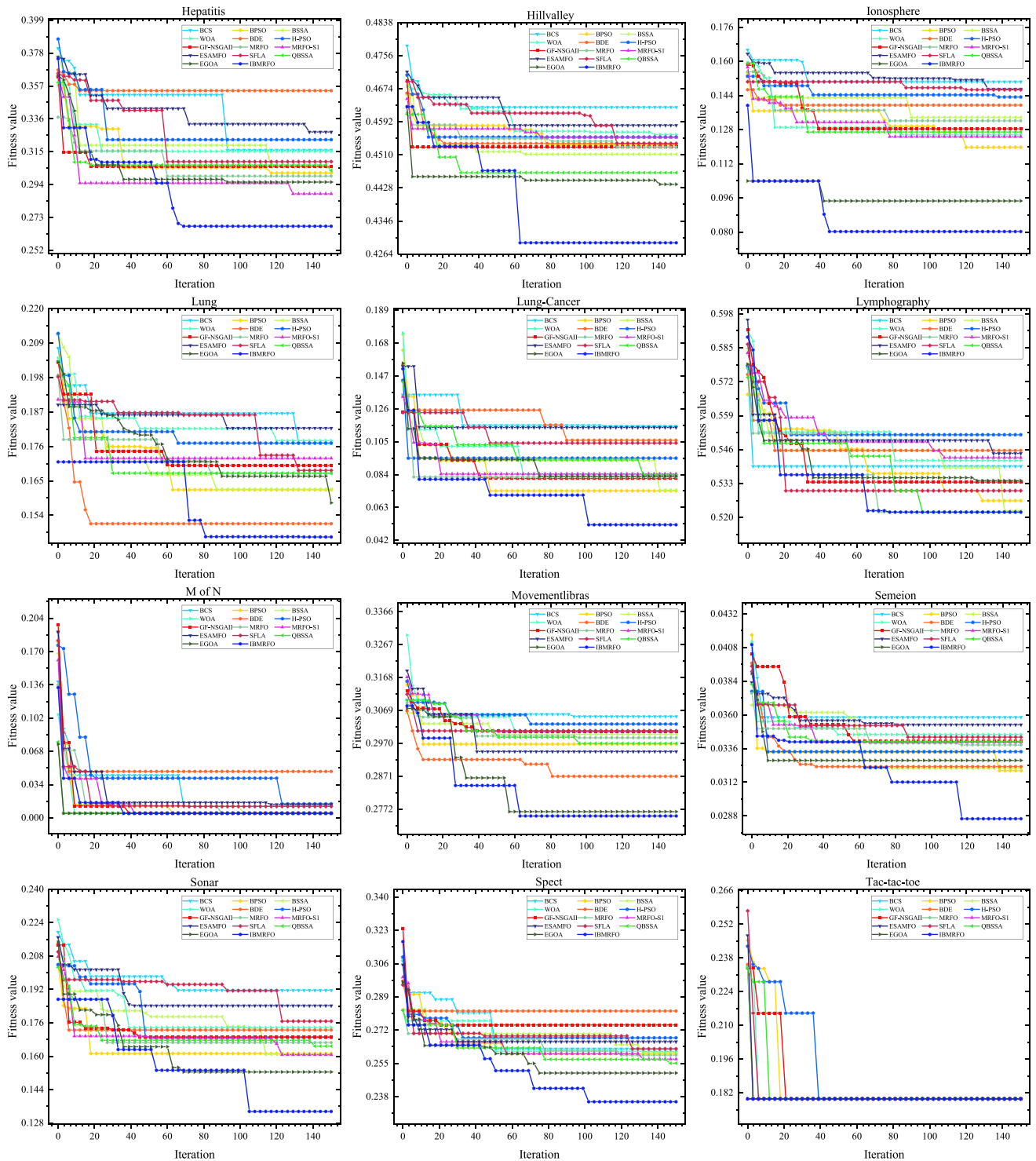


Fig. 9. Rate of convergence derived from diverse algorithms (Part 2).

In terms of the standard deviation of the average number of selected features, BPSO is ahead in 16 datasets, indicating its consistency in feature selection. It is important to highlight that although IBMRFO is slightly less effective than GF-NSGAIL in minimizing the selected features, it still secures the second place in terms of performance. This is further illustrated in Fig. 11(a), which clearly shows IBMRFO's competitive advantage in feature subset size over all algorithms except GF-NSGAIL.

4.2.4. Features selection F1 weighted score

The average F1-weighted scores collected by different algorithms are presented in the Table 7. IBMRFO achieves the best average F1-weighted values in 15 datasets and the lowest average standard deviation of F1-weighted values in 11 datasets. The F1-weighted scores of each algorithm for 30 datasets are shown in Fig. 11(b), where the edge of Fig. 11(b) corresponds to an F1-weighted score of 1.0, with values closer to the Edge indicate superior algorithmic F1-weighted

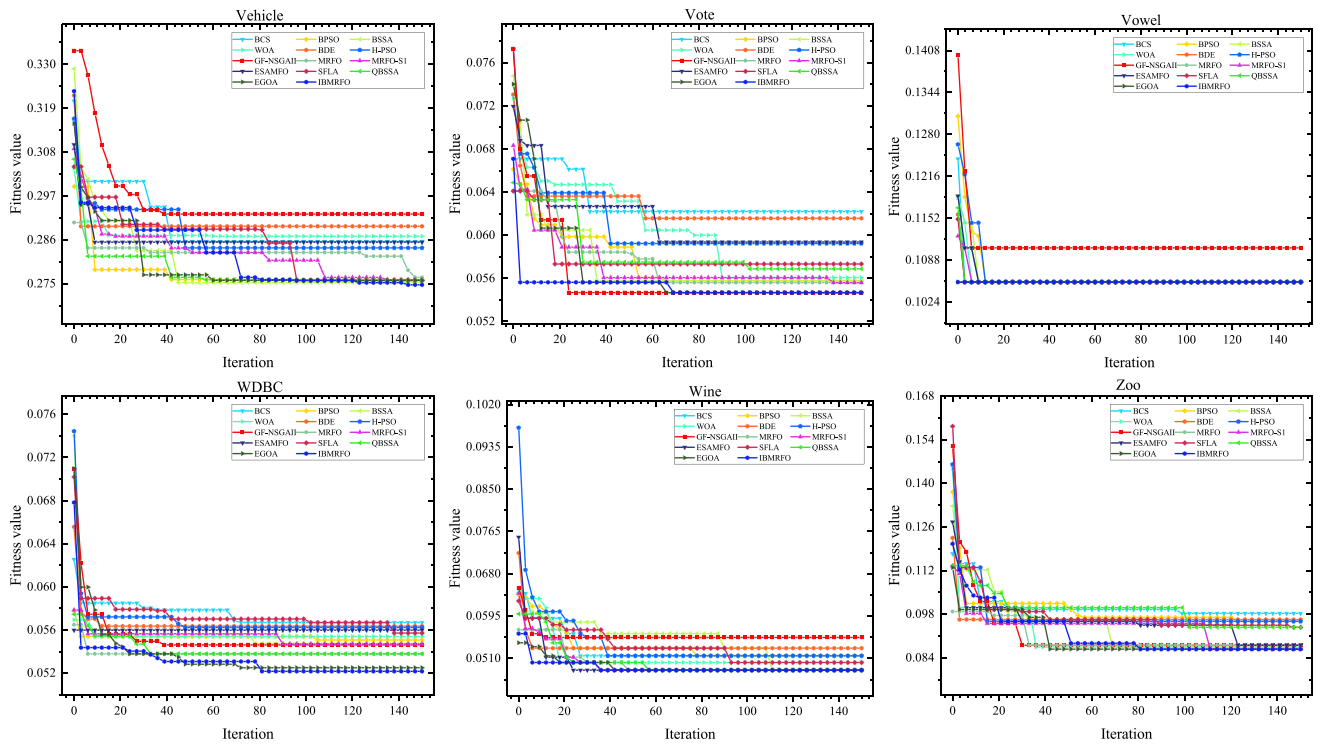


Fig. 10. Rate of convergence derived from diverse algorithms (Part 3).

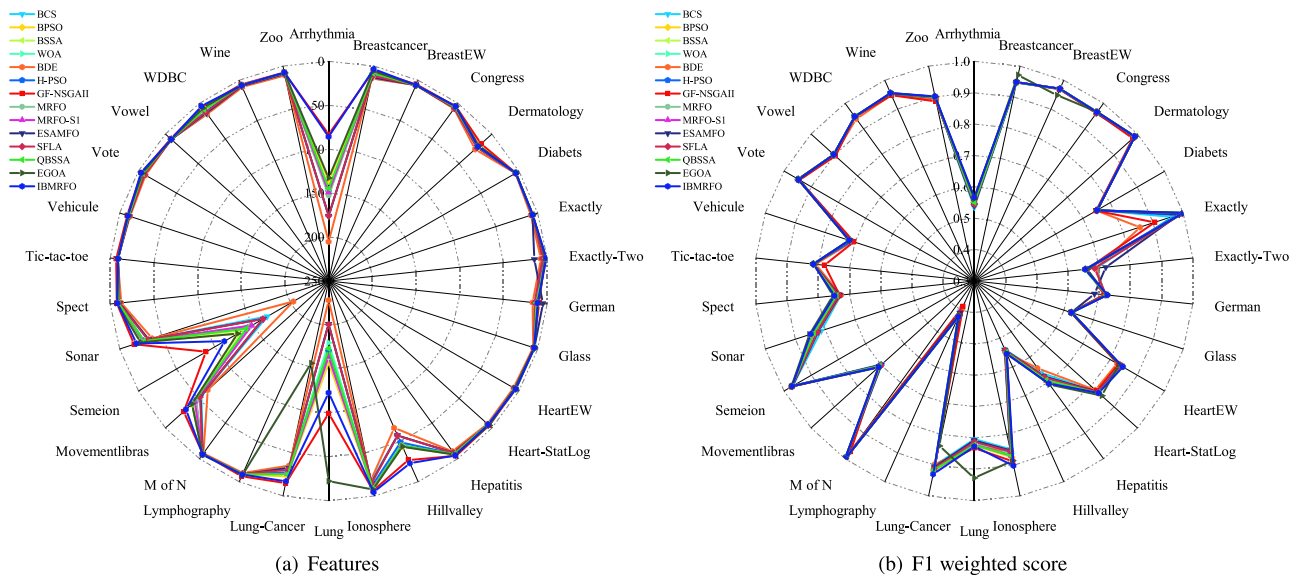


Fig. 11. The average number of selected features and F1 weighted score radar chart of different algorithms.

score. The IBMRFO algorithm, characterized by its proximity to the 1.0 mark on most axes, consistently has a high average F1-weighted score on most datasets. This means that the proposed IBMRFO algorithm has an advantage in F1-weighted evaluation.

4.2.5. CPU execution time

Furthermore, the average CPU time and standard deviation achieved by different algorithms for each data set are given in the Table 8. The Table 8 clearly indicates that EGOA has a significant advantage in terms of the average execution time of the algorithm, and ESAMFO has an advantage in terms of the average execution standard deviation

of the algorithm. Although IBMRFO takes more time to achieve convergence, it offers greater advantages in classification accuracy, number of selected features and F1-weighted score.

After carefully evaluating the fitness function values, classification accuracy, F1 weighted score, number of selected features, and running time of the algorithm, IBMRFO has clearly emerged as a leading algorithm. While the IBMRFO algorithm presents certain limitations in terms of running time, it exhibits pronounced advantages across other key performance indicators. Specifically, IBMRFO achieves the best average fitness function values in 17 of the datasets according to Table 4, outperforms in classification accuracy in 16 datasets according to Table 5, and outperforms the best average F1 weighted scores in 15 datasets according to Table 7. Meanwhile, IBMRFO achieves the

Table 7
Average f1 weighted scores achieved by different algorithms.

	BCS $F1_w \pm \text{std}$	BPSO $F1_w \pm \text{std}$	BSSA $F1_w \pm \text{std}$	WOA $F1_w \pm \text{std}$	BDE $F1_w \pm \text{std}$	H-PSO $F1_w \pm \text{std}$	GF-NSGAI $F1_w \pm \text{std}$	MRFO $F1_w \pm \text{std}$	MRFO-S1 $F1_w \pm \text{std}$	ESAMFO $F1_w \pm \text{std}$	SFLA $F1_w \pm \text{std}$	QBSA $F1_w \pm \text{std}$	EGOA $F1_w \pm \text{std}$	IBMRFO $F1_w \pm \text{std}$
Arrhythmia	0.5350 $\pm$ 0.0051	0.5553 $\pm$ 0.0050	0.5517 $\pm$ 0.0066	0.5446 $\pm$ 0.0050	0.5476 $\pm$ 0.0098	0.5442 $\pm$ 0.0080	0.5649 $\pm$ 0.0174	0.5489 $\pm$ 0.0065	0.5501 $\pm$ 0.0060	0.5406 $\pm$ 0.0057	0.5441 $\pm$ 0.0056	0.5507 $\pm$ 0.0078	0.5622 $\pm$ 0.0067	0.5667 $\pm$ 0.0123
Breastcancer	0.9476 $\pm$ 0.0005	0.9478 $\pm$ 0.0002	0.9477 $\pm$ 0.0003	0.9476 $\pm$ 0.0004	0.9462 $\pm$ 0.0035	0.9476 $\pm$ 0.0005	0.9477 $\pm$ 0.0005	0.9477 $\pm$ 0.0003	0.9477 $\pm$ 0.0003	0.9478 $\pm$ 0.0003	0.9478 $\pm$ 0.0003	0.9476 $\pm$ 0.0004	0.9478 $\pm$ 0.0002	0.9479 $\pm$ 0.0001
BreastEW	0.9706 $\pm$ 0.0003	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9695 $\pm$ 0.0008	0.9706 $\pm$ 0.0003	0.9686 $\pm$ 0.0019	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0	0.9706 $\pm$ 0.0
Congress	0.9613 $\pm$ 0.0024	0.9663 $\pm$ 0.0010	0.9658 $\pm$ 0.0015	0.9642 $\pm$ 0.0021	0.9604 $\pm$ 0.0041	0.9629 $\pm$ 0.0025	0.9629 $\pm$ 0.0032	0.9658 $\pm$ 0.0016	0.9656 $\pm$ 0.0014	0.9637 $\pm$ 0.0028	0.9645 $\pm$ 0.0021	0.9656 $\pm$ 0.0018	0.9662 $\pm$ 0.0007	0.9657 $\pm$ 0.0013
Dermatology	0.9866 $\pm$ 0.0014	0.9868 $\pm$ 0.0013	0.9889 $\pm$ 0.0009	0.9879 $\pm$ 0.0015	0.9870 $\pm$ 0.0025	0.9854 $\pm$ 0.0014	0.9797 $\pm$ 0.0081	0.9883 $\pm$ 0.0012	0.9882 $\pm$ 0.0011	0.9879 $\pm$ 0.0013	0.9880 $\pm$ 0.0012	0.9876 $\pm$ 0.0011	0.9888 $\pm$ 0.0014	0.9882 $\pm$ 0.0011
Diabets	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7489 $\pm$ 0.0025	0.7512 $\pm$ 0.0	0.7479 $\pm$ 0.0069	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0	0.7512 $\pm$ 0.0
Exactly	0.9621 $\pm$ 0.0272	0.9990 $\pm$ 0.0054	0.9979 $\pm$ 0.0082	0.9917 $\pm$ 0.0140	0.8550 $\pm$ 0.0811	0.9796 $\pm$ 0.0172	0.9048 $\pm$ 0.1356	1.0000 $\pm$ 0.0	1.0000 $\pm$ 0.0	0.9876 $\pm$ 0.0155	0.9970 $\pm$ 0.0090	1.0000 $\pm$ 0.0	1.0000 $\pm$ 0.0	1.0000 $\pm$ 0.0
Exactly-two	0.6924 $\pm$ 0.0272	0.6551 $\pm$ 0.0007	0.6555 $\pm$ 0.0050	0.6592 $\pm$ 0.0155	0.6909 $\pm$ 0.0229	0.6609 $\pm$ 0.0150	0.6601 $\pm$ 0.0156	0.6562 $\pm$ 0.0048	0.6557 $\pm$ 0.0046	0.6852 $\pm$ 0.0275	0.6851 $\pm$ 0.0296	0.6542 $\pm$ 0.0019	0.6602 $\pm$ 0.0161	0.6551 $\pm$ 0.0
German	0.7184 $\pm$ 0.0031	0.7241 $\pm$ 0.0039	0.7240 $\pm$ 0.0036	0.7231 $\pm$ 0.0037	0.7162 $\pm$ 0.0072	0.7191 $\pm$ 0.0040	0.7163 $\pm$ 0.0080	0.7243 $\pm$ 0.0028	0.7246 $\pm$ 0.0042	0.7214 $\pm$ 0.0033	0.7223 $\pm$ 0.0034	0.7239 $\pm$ 0.0033	0.7255 $\pm$ 0.0037	0.7265 $\pm$ 0.0031
Glass	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6247 $\pm$ 0.0044	0.6255 $\pm$ 0.0	0.6236 $\pm$ 0.0055	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6255 $\pm$ 0.0	0.6251 $\pm$ 0.0013	0.6255 $\pm$ 0.0
HeartEW	0.8363 $\pm$ 0.0046	0.8459 $\pm$ 0.0012	0.8442 $\pm$ 0.0026	0.8431 $\pm$ 0.0035	0.8262 $\pm$ 0.0265	0.8383 $\pm$ 0.0048	0.8304 $\pm$ 0.0246	0.8457 $\pm$ 0.0015	0.8454 $\pm$ 0.0017	0.8425 $\pm$ 0.0034	0.8419 $\pm$ 0.0032	0.8456 $\pm$ 0.0016	0.8325 $\pm$ 0.0031	0.8462 $\pm$ 0.0009
Heart-StatLog	0.8267 $\pm$ 0.0037	0.8335 $\pm$ 0.0007	0.8330 $\pm$ 0.0018	0.8308 $\pm$ 0.0036	0.8185 $\pm$ 0.0084	0.8296 $\pm$ 0.0038	0.8242 $\pm$ 0.0076	0.8327 $\pm$ 0.0022	0.8329 $\pm$ 0.0018	0.8294 $\pm$ 0.0044	0.8302 $\pm$ 0.0043	0.8331 $\pm$ 0.0017	0.8461 $\pm$ 0.0011	0.8336 $\pm$ 0.0005
Hepatitis	0.6625 $\pm$ 0.0124	0.6973 $\pm$ 0.0120	0.6910 $\pm$ 0.0115	0.6816 $\pm$ 0.0105	0.6433 $\pm$ 0.0194	0.6758 $\pm$ 0.0109	0.6822 $\pm$ 0.0201	0.6912 $\pm$ 0.0122	0.6923 $\pm$ 0.0111	0.6741 $\pm$ 0.0130	0.6794 $\pm$ 0.0106	0.6905 $\pm$ 0.0093	0.7080 $\pm$ 0.0143	0.7039 $\pm$ 0.0147
Hillvalley	0.5392 $\pm$ 0.0026	0.5486 $\pm$ 0.0028	0.5475 $\pm$ 0.0033	0.5442 $\pm$ 0.0028	0.5505 $\pm$ 0.0046	0.5422 $\pm$ 0.0023	0.5533 $\pm$ 0.0096	0.5455 $\pm$ 0.0024	0.5458 $\pm$ 0.0038	0.5406 $\pm$ 0.0028	0.5415 $\pm$ 0.0023	0.5480 $\pm$ 0.0031	0.5528 $\pm$ 0.0034	0.5545 $\pm$ 0.0046
Ionosphere	0.8475 $\pm$ 0.0045	0.8738 $\pm$ 0.0068	0.8700 $\pm$ 0.0089	0.8626 $\pm$ 0.0062	0.8556 $\pm$ 0.0102	0.8564 $\pm$ 0.0063	0.8881 $\pm$ 0.0178	0.8717 $\pm$ 0.0080	0.8686 $\pm$ 0.0071	0.8517 $\pm$ 0.0043	0.8543 $\pm$ 0.0057	0.8690 $\pm$ 0.0073	0.8857 $\pm$ 0.0087	0.9014 $\pm$ 0.0100
Lung	0.8039 $\pm$ 0.0078	0.8239 $\pm$ 0.0085	0.8205 $\pm$ 0.0085	0.8142 $\pm$ 0.0095	0.8253 $\pm$ 0.0127	0.8090 $\pm$ 0.0059	0.8320 $\pm$ 0.0202	0.8209 $\pm$ 0.0096	0.8212 $\pm$ 0.0088	0.8081 $\pm$ 0.0072	0.8128 $\pm$ 0.0068	0.8221 $\pm$ 0.0094	0.9274 $\pm$ 0.0127	0.8274 $\pm$ 0.0097
Lung-Cancer	0.9000 $\pm$ 0.0108	0.9192 $\pm$ 0.0066	0.9163 $\pm$ 0.0081	0.9086 $\pm$ 0.0078	0.8959 $\pm$ 0.0159	0.9060 $\pm$ 0.0092	0.9125 $\pm$ 0.0169	0.9157 $\pm$ 0.0085	0.9140 $\pm$ 0.0089	0.9052 $\pm$ 0.0083	0.9099 $\pm$ 0.0092	0.9157 $\pm$ 0.0105	0.8370 $\pm$ 0.0097	0.9295 $\pm$ 0.0121
Lymphography	0.4064 $\pm$ 0.0124	0.4238 $\pm$ 0.0105	0.4271 $\pm$ 0.0080	0.4177 $\pm$ 0.0113	0.3942 $\pm$ 0.0179	0.4097 $\pm$ 0.0100	0.3886 $\pm$ 0.0251	0.4242 $\pm$ 0.0113	0.4232 $\pm$ 0.0097	0.4116 $\pm$ 0.0106	0.4202 $\pm$ 0.0120	0.4256 $\pm$ 0.0091	0.4203 $\pm$ 0.0176	0.4259 $\pm$ 0.0103
M of N	0.9917 $\pm$ 0.0096	1.0000 $\pm$ 0.0	1.0000 $\pm$ 0.0	0.9993 $\pm$ 0.0020	0.9713 $\pm$ 0.0255	0.9923 $\pm$ 0.0084	0.9725 $\pm$ 0.0502	0.9998 $\pm$ 0.0012	0.9998 $\pm$ 0.0012	0.9976 $\pm$ 0.0033	0.9983 $\pm$ 0.0029	0.9998 $\pm$ 0.0012	1.0000 $\pm$ 0.0	0.9998 $\pm$ 0.0012
Movementlibras	0.6966 $\pm$ 0.0039	0.7049 $\pm$ 0.0038	0.7051 $\pm$ 0.0043	0.7010 $\pm$ 0.0047	0.7057 $\pm$ 0.0056	0.6984 $\pm$ 0.0042	0.7099 $\pm$ 0.0088	0.7036 $\pm$ 0.0042	0.7043 $\pm$ 0.0048	0.6995 $\pm$ 0.0039	0.7009 $\pm$ 0.0042	0.7046 $\pm$ 0.0043	0.7112 $\pm$ 0.0053	0.7096 $\pm$ 0.0065
Semeion	0.9685 $\pm$ 0.0008	0.9707 $\pm$ 0.0009	0.9704 $\pm$ 0.0015	0.9690 $\pm$ 0.0011	0.9736 $\pm$ 0.0015	0.9689 $\pm$ 0.0011	0.9714 $\pm$ 0.0027	0.9702 $\pm$ 0.0011	0.9704 $\pm$ 0.0009	0.9698 $\pm$ 0.0011	0.9701 $\pm$ 0.0009	0.9702 $\pm$ 0.0009	0.9724 $\pm$ 0.0014	0.9706 $\pm$ 0.0016
Sonar	0.8125 $\pm$ 0.0057	0.8422 $\pm$ 0.0067	0.8367 $\pm$ 0.0097	0.8299 $\pm$ 0.0082	0.8281 $\pm$ 0.0144	0.8249 $\pm$ 0.0085	0.8387 $\pm$ 0.0175	0.8378 $\pm$ 0.0115	0.8378 $\pm$ 0.0083	0.8233 $\pm$ 0.0081	0.8248 $\pm$ 0.0083	0.8374 $\pm$ 0.0072	0.8526 $\pm$ 0.0095	0.8474 $\pm$ 0.0108
Spect	0.7272 $\pm$ 0.0067	0.7436 $\pm$ 0.0045	0.7419 $\pm$ 0.0055	0.7388 $\pm$ 0.0070	0.7270 $\pm$ 0.0104	0.7320 $\pm$ 0.0059	0.7312 $\pm$ 0.0129	0.7405 $\pm$ 0.0044	0.7407 $\pm$ 0.0056	0.7320 $\pm$ 0.0058	0.7338 $\pm$ 0.0058	0.7412 $\pm$ 0.0054	0.7524 $\pm$ 0.0052	0.7478 $\pm$ 0.0061
Tic-tac-toe	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8083 $\pm$ 0.0169	0.8157 $\pm$ 0.0	0.7794 $\pm$ 0.0273	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0	0.8122 $\pm$ 0.0133	0.8157 $\pm$ 0.0	0.8157 $\pm$ 0.0
Vehicule	0.7121 $\pm$ 0.0053	0.7179 $\pm$ 0.0031	0.7202 $\pm$ 0.0021	0.7172 $\pm$ 0.0044	0.7076 $\pm$ 0.0064	0.7102 $\pm$ 0.0041	0.7039 $\pm$ 0.0115	0.7190 $\pm$ 0.0034	0.7186 $\pm$ 0.0036	0.7167 $\pm$ 0.0046	0.7175 $\pm$ 0.0035	0.7185 $\pm$ 0.0035	0.7197 $\pm$ 0.0032	0.7191 $\pm$ 0.0034
Vote	0.9444 $\pm$ 0.0019	0.9473 $\pm$ 0.0015	0.9477 $\pm$ 0.0014	0.9476 $\pm$ 0.0012	0.9438 $\pm$ 0.0044	0.9454 $\pm$ 0.0026	0.9435 $\pm$ 0.0038	0.9474 $\pm$ 0.0016	0.9479 $\pm$ 0.0014	0.9463 $\pm$ 0.0021	0.9477 $\pm$ 0.0011	0.9477 $\pm$ 0.0015	0.9478 $\pm$ 0.0010	0.9468 $\pm$ 0.0015
Vowel	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9003 $\pm$ 0.0037	0.9027 $\pm$ 0.0	0.8955 $\pm$ 0.0072	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0	0.9027 $\pm$ 0.0
WDBC	0.9473 $\pm$ 0.0007	0.9477 $\pm$ 0.0003	0.9476 $\pm$ 0.0003	0.9476 $\pm$ 0.0004	0.9391 $\pm$ 0.0081	0.9475 $\pm$ 0.0006	0.9465 $\pm$ 0.0037	0.9477 $\pm$ 0.0003	0.9477 $\pm$ 0.0003	0.9477 $\pm$ 0.0003	0.9477 $\pm$ 0.0004	0.9477 $\pm$ 0.0004	0.9479 $\pm$ 0.0001	0.9478 $\pm$ 0.0002
Wine	0.9534 $\pm$ 0.0024	0.9548 $\pm$ 0.0014	0.9546 $\pm$ 0.0015	0.9547 $\pm$ 0.0016	0.9488 $\pm$ 0.0028	0.9527 $\pm$ 0.0023	0.9476 $\pm$ 0.0039	0.9552 $\pm$ 0.0007	0.9552 $\pm$ 0.0007	0.9549 $\pm$ 0.0012	0.9545 $\pm$ 0.0016	0.9549 $\pm$ 0.0012	0.9537 $\pm$ 0.0023	0.9553 $\pm$ 0.0
Zoo	0.8940 $\pm$ 0.0056	0.8995 $\pm$ 0.0045	0.8992 $\pm$ 0.0050	0.8984 $\pm$ 0.0059	0.8902 $\pm$ 0.0103	0.8970 $\pm$ 0.0065	0.8857 $\pm$ 0.0133	0.9011 $\pm$ 0.0042	0.9001 $\pm$ 0.0040	0.8974 $\pm$ 0.0055	0.8976 $\pm$ 0.0051	0.8986 $\pm$ 0.0044	0.9006 $\pm$ 0.0023	0.9013 $\pm$ 0.0035
Mean	0.8113 $\pm$ 0.0056	0.8198 $\pm$ 0.0031	0.8190 $\pm$ 0.0037	0.8163 $\pm$ 0.0045	0.8067 $\pm$ 0.0121	0.8134 $\pm$ 0.0049	0.8105 $\pm$ 0.0169	0.8189 $\pm$ 0.0035	0.8188 $\pm$ 0.0034	0.8149 $\pm$ 0.0049	0.8166 $\pm$ 0.0046	0.8189 $\pm$ 0.0033	0.8227 $\pm$ 0.0048	0.8228 $\pm$ 0.0039
Outperform	6(6)	7(8)	9(9)	5(6										

Table 8
Average CPU time occupied by different algorithms(/s).

	BCS time ± std	BPSO time ± std	BSSA time ± std	WOA time ± std	BDE time ± std	H-PSO time ± std	GF-NSGAII time ± std	MRFO time ± std	MRFO-S1 time ± std	ESAMFO time ± std	SFLA time ± std	QBSSA time ± std	EGOA time ± std	IBMRFO time ± std
Arrhythmia	209.1 ± 2.7	188.1 ± 3.3	236.9 ± 3.4	222.0 ± 5.3	143.1 ± 1.1	161.7 ± 18.4	171.0 ± 6.3	399.5 ± 34.9	405.5 ± 13.0	137.4 ± 2.0	122.6 ± 1.6	537.2 ± 31.3	161. ± 22.1	597.9 ± 14.9
Breastcancer	191.0 ± 5.8	152.7 ± 5.1	180.8 ± 8.2	190.5 ± 9.2	128.1 ± 18.2	160.3 ± 8.0	111.8 ± 4.1	369.2 ± 9.1	378.5 ± 10.2	128.0 ± 10.4	140.1 ± 1.1	419.7 ± 24.5	100.8 ± 2.8	530.8 ± 11.1
BreastEW	138.3 ± 19.1	137.8 ± 31.6	147.0 ± 24.3	140.5 ± 19.8	124.6 ± 1.4	129.5 ± 17.7	133.3 ± 10.0	270.7 ± 16.1	277.0 ± 35.1	121.6 ± 0.6	122.4 ± 4.0	365.9 ± 26.8	89.4 ± 3.3	405.3 ± 23.5
Congress	110.0 ± 9.9	102.3 ± 10.9	101.1 ± 11.0	106.5 ± 14.3	99.2 ± 18.4	101.3 ± 12.2	92.3 ± 13.6	225.9 ± 24.5	214.0 ± 27.0	103.3 ± 8.2	100.6 ± 1.3	318.0 ± 23.1	99.1 ± 27.3	346.8 ± 22.7
Dermatology	154.3 ± 3.1	152.4 ± 5.5	155.2 ± 3.4	153.9 ± 5.0	140.6 ± 4.0	137.0 ± 5.5	109.3 ± 14.1	300.2 ± 8.5	291.3 ± 5.5	132.0 ± 2.0	128.8 ± 4.8	439.0 ± 24.5	107.0 ± 9.2	446.0 ± 12.1
Diabetes	142.6 ± 8.6	143.1 ± 20.3	146.5 ± 14.4	146.8 ± 14.8	143.8 ± 11.7	133.8 ± 8.7	129.7 ± 14.1	290.9 ± 18.6	292.5 ± 20.8	142.2 ± 6.2	141.7 ± 10.1	418.7 ± 27.4	109.6 ± 7.5	433.2 ± 37.1
Exactly	215.0 ± 36.6	195.0 ± 25.1	201.9 ± 22.9	196.3 ± 19.7	206.1 ± 21.7	189.4 ± 19.9	169.6 ± 18.1	403.7 ± 22.6	394.3 ± 18.4	199.7 ± 9.5	194.3 ± 22.9	603.2 ± 42.6	165.4 ± 36.5	595.2 ± 33.7
Exactly-Two	217.1 ± 23.0	197.5 ± 28.0	193.3 ± 12.2	201.9 ± 24.2	198.8 ± 21.6	184.4 ± 10.1	157.0 ± 13.8	405.1 ± 28.7	398.0 ± 24.5	213.5 ± 9.1	202.2 ± 18.9	570.4 ± 43.0	138.3 ± 6.0	594.1 ± 34.4
German	294.7 ± 11.8	222.6 ± 16.4	252.8 ± 9.3	289.3 ± 10.8	229.1 ± 26.0	188.2 ± 1.1	160.3 ± 9.1	554.9 ± 17.6	541.6 ± 12.8	200.2 ± 13.3	185.3 ± 2.8	562.1 ± 6.9	136.4 ± 4.6	877.0 ± 25.0
Glass	75.5 ± 21.0	73.1 ± 18.1	73.4 ± 17.7	68.1 ± 9.3	65.1 ± 7.2	68.8 ± 18.1	68.4 ± 15.1	140.9 ± 15.7	138.4 ± 29.4	71.0 ± 14.2	73.1 ± 18.5	198.7 ± 29.3	76.8 ± 16.5	207.8 ± 22.0
HeartEW	75.7 ± 21.5	78.4 ± 17.6	83.5 ± 19.8	76.0 ± 14.9	79.3 ± 14.9	75.3 ± 14.4	75.2 ± 11.8	145.2 ± 10.0	149.8 ± 19.4	73.6 ± 10.2	78.2 ± 13.1	229.6 ± 23.7	58.9 ± 14.1	224.1 ± 20.6
Heart-StatLog	82.7 ± 20.8	84.0 ± 16.9	82.2 ± 16.8	84.9 ± 14.3	84.5 ± 12.7	82.0 ± 15.5	79.9 ± 16.1	164.1 ± 19.8	164.2 ± 16.9	84.1 ± 16.9	87.4 ± 21.3	241.6 ± 18.9	54.7 ± 10.3	246.8 ± 27.5
Hepatitis	75.0 ± 4.3	59.9 ± 12.9	55.6 ± 3.8	66.6 ± 6.8	51.2 ± 0.7	50.5 ± 1.4	50.5 ± 0.7	142.5 ± 13.3	140.7 ± 9.9	46.6 ± 0.1	45.9 ± 0.3	179.9 ± 9.3	67.7 ± 9.7	221.3 ± 8.3
Hillvalley	214.7 ± 6.8	209.6 ± 3.7	197.0 ± 5.0	222.1 ± 3.2	147.5 ± 1.1	145.1 ± 4.5	142.4 ± 10.2	454.3 ± 10.5	463.8 ± 13.5	138.6 ± 9.1	129.0 ± 0.6	490.6 ± 52.9	143.8 ± 16.5	669.4 ± 19.0
Ionosphere	134.4 ± 4.4	126.5 ± 2.9	119.8 ± 3.6	145.1 ± 2.5	90.7 ± 5.1	95.4 ± 0.5	76.9 ± 3.5	290.2 ± 6.2	301.4 ± 5.7	93.0 ± 3.6	81.8 ± 0.5	286.6 ± 3.7	78.5 ± 12.2	415.6 ± 10.3
Lung	82.2 ± 2.7	85.8 ± 2.1	140.4 ± 4.6	98.5 ± 2.2	82.4 ± 2.9	90.8 ± 2.1	85.5 ± 10.4	180.0 ± 8.3	172.4 ± 2.5	85.9 ± 1.4	86.3 ± 1.6	249.3 ± 6.1	30.9 ± 9.4	255.2 ± 6.0
Lung-Cancer	36.7 ± 9.6	34.8 ± 7.7	40.9 ± 12.7	41.9 ± 11.8	39.0 ± 12.0	36.3 ± 6.1	45.4 ± 13.9	72.9 ± 14.1	78.7 ± 18.4	40.8 ± 12.3	37.5 ± 9.7	116.4 ± 24.9	79.3 ± 3.6	112.8 ± 20.2
Lymphography	70.3 ± 5.5	61.2 ± 10.5	65.8 ± 5.7	70.3 ± 6.4	70.1 ± 23.1	64.8 ± 7.6	65.9 ± 20.9	133.3 ± 11.4	135.1 ± 9.1	74.1 ± 5.8	70.8 ± 5.6	193.7 ± 17.5	45.4 ± 8.8	206.2 ± 10.4
M of N	209.3 ± 28.3	200.1 ± 22.8	198.8 ± 14.9	200.3 ± 21.2	226.0 ± 35.4	197.2 ± 22.5	188.7 ± 31.5	403.9 ± 21.6	394.4 ± 34.4	200.2 ± 13.2	193.3 ± 30.8	574.2 ± 49.3	165.9 ± 42.4	606.6 ± 70.1
Movementlibras	154.7 ± 4.2	157.6 ± 2.6	166.2 ± 5.0	156.1 ± 4.3	145.8 ± 2.4	135.7 ± 2.0	141.7 ± 8.8	314.2 ± 7.8	302.2 ± 6.1	135.0 ± 3.7	129.8 ± 6.7	455.4 ± 10.8	92.9 ± 11.8	460.3 ± 12.6
Sonar	648.0 ± 13.1	624.5 ± 22.6	642.8 ± 13.9	776.1 ± 15.5	536.3 ± 51.7	546.4 ± 2.5	405.1 ± 13.0	1535.9 ± 30.0	1334.0 ± 48.8	550.5 ± 3.0	497.5 ± 11.5	1741.1 ± 189.6	468.2 ± 95.2	2249.3 ± 38.3
Spect	118.3 ± 3.1	113.6 ± 3.2	120.7 ± 2.6	119.1 ± 4.7	107.3 ± 1.9	93.2 ± 1.1	83.5 ± 12.5	229.4 ± 4.4	227.7 ± 5.7	87.1 ± 1.7	100.1 ± 2.6	303.8 ± 8.8	57.1 ± 3.4	343.5 ± 8.7
Tic-Tac-toe	111.3 ± 5.0	92.0 ± 9.6	104.5 ± 6.1	103.1 ± 4.9	87.2 ± 18.9	92.4 ± 4.5	70.0 ± 9.3	202.2 ± 11.7	195.5 ± 6.4	93.6 ± 1.7	89.1 ± 3.1	284.2 ± 7.1	88.9 ± 18.7	300.6 ± 10.5
Vehicule	195.9 ± 22.5	195.9 ± 23.4	195.9 ± 20.1	191.8 ± 25.5	194.1 ± 15.2	176.1 ± 19.0	175.1 ± 19.4	371.9 ± 23.3	372.7 ± 25.2	188.8 ± 11.6	182.9 ± 14.8	563.5 ± 35.7	146.2 ± 4.4	566.8 ± 45.5
Vowel	201.3 ± 10.8	184.2 ± 22.4	188.1 ± 12.0	196.0 ± 11.8	166.1 ± 11.2	158.6 ± 12.7	149.2 ± 3.7	385.7 ± 19.4	377.6 ± 21.7	156.9 ± 2.2	147.7 ± 2.3	513.8 ± 26.0	150.4 ± 37.6	592.1 ± 20.4
Vote	87.0 ± 15.7	81.9 ± 7.8	80.9 ± 14.3	89.2 ± 13.7	70.2 ± 2.7	70.4 ± 1.7	68.8 ± 2.5	164.1 ± 14.1	167.1 ± 18.0	72.1 ± 0.5	72.9 ± 0.9	237.9 ± 24.5	56.7 ± 3.3	275.8 ± 20.1
Wine	168.8 ± 12.6	167.3 ± 15.2	169.2 ± 12.6	185.8 ± 19.9	174.1 ± 23.0	152.8 ± 1.4	157.0 ± 3.5	360.4 ± 37.1	353.9 ± 34.5	150.1 ± 9.3	162.6 ± 2.1	495.2 ± 27.0	131.3 ± 5.5	527.0 ± 42.5
WDBC	192.2 ± 4.1	167.1 ± 4.8	184.4 ± 6.0	187.9 ± 6.1	166.7 ± 27.6	180.9 ± 3.7	126.3 ± 14.8	376.0 ± 20.5	370.1 ± 13.1	186.5 ± 3.9	186.6 ± 7.3	569.7 ± 15.6	137.1 ± 27.5	545.5 ± 11.9
WineC	61.0 ± 12.7	73.6 ± 24.0	66.0 ± 15.8	64.2 ± 8.5	69.9 ± 26.7	65.2 ± 10.1	60.7 ± 8.4	130.4 ± 29.5	125.6 ± 24.6	62.4 ± 12.5	59.4 ± 10.6	175.8 ± 13.7	44.8 ± 3.3	195.7 ± 38.3
Zoo	56.0 ± 14.1	56.0 ± 18.7	55.2 ± 13.6	53.8 ± 14.1	48.2 ± 12.5	52.6 ± 11.6	49.4 ± 7.6	98.0 ± 18.2	96.2 ± 16.9	47.8 ± 8.3	51.5 ± 12.7	143.1 ± 18.4	36.9 ± 1.0	146.5 ± 13.5
Mean	157.4 ± 12.1	147.3 ± 13.9	154.9 ± 11.2	161.5 ± 11.5	137.2 ± 14.4	133.9 ± 8.8	120.0 ± 11.4	317.2 ± 17.6	308.5 ± 18.2	133.88 ± 6.89	130.04 ± 8.54	415.95 ± 28.76	110.65 ± 15.82	473.2 ± 23.0
Outperform	1(4)	2(1)	0(1)	0(0)	3(7)	2(7)	5(4)	0(1)	0(0)	0(11)	4(2)	0(0)	18(7)	0(0)

best fitness function convergence curve on 15 datasets and achieves the same fitness function convergence curve with different algorithms on others according to Figs. 8, 9 and 10. Nevertheless, in the aspect of the number of selected features, IBMRFO demonstrated superiority in 11 datasets. In summary, these experimental results confirm the superior performance of the IBMRFO algorithm in the field of feature selection.

4.3. Effectiveness of the improved factors

In this section, a series of experiments are conducted to evaluate the effectiveness of the proposed factors in IBMRFO. The aim is to find out whether these factors contribute to improving the performance of the traditional MRFO algorithm. To achieve this, we compare the performance of the traditional MRFO algorithm with three variants: MRFO with CTM mechanism (T-MRFO), MRFO with ASF factor (S-MRFO), and the fusion of MRFO with CTM mechanism and ASF factor (IBMRFO). These algorithms are used to solve the feature selection problem formulated in this study by using a set of carefully selected datasets, namely Arrhythmia, Diabetes, HeartEW, Hillvalley, Lung and Tic-Tac-Toe. The numerical results generated by these above methods are shown in the Table 9. In addition, the convergence rates of the various improvement factors used in the optimization processes are shown in Fig. 12. The remaining findings are explained in detail in the subsequent discussion.

4.3.1. Effectiveness of the chaotic tent map (CTM) mechanism

As shown in Table 9, T-MRFO exhibits improved performance over conventional MRFO on 4 datasets, as evidenced by improvements in the average fitness function value, average selected number of features, and algorithm execution time. In addition, it excels in average precision and average F1 weighted score on 5 datasets. This consistent improvement across different metrics highlights the effectiveness of T-MRFO and demonstrates its ability to effectively optimize the fitness function while achieving high accuracy and maintaining computational efficiency.

The reduction in the average number of selected features indicates that T-MRFO is effective in feature reduction, which is essential for simplifying models, reducing overfitting, and potentially speeding up algorithm execution. The balance between the number of selected features and the fitness value also indicates the ability of T-MRFO to maintain or improve performance even with a reduced feature set, indicating a more intelligent feature selection process. Therefore, the proposed CTM mechanism can improve the feature selection performance of the traditional MRFO algorithm.

4.3.2. Effectiveness of the adaptive somersault factor (ASF)

Table 9 illustrates that, compared to conventional MRFO, S-MRFO exhibits superior performance in 4 datasets in terms of average fitness function value and average F1 weighted score. Additionally, it outshines in 5 datasets regarding average accuracy. However, S-MRFO falls slightly short in average algorithm execution time and the average number of selected features. This indicates that while S-MRFO excels in achieving higher accuracy and improved fitness function values, these advantages come at the cost of longer computational time and potentially a more complex set of features.

Table 9 shows that S-MRFO improved compared to conventional MRFO in 4 datasets, especially in terms of average fitness function value and average F1 weighted score. Furthermore, it outperforms average accuracy in 5 datasets. However, S-MRFO is slightly behind in terms of the average execution time of the algorithm and the average number of selected features. This trend suggests that although S-MRFO is capable of achieving higher accuracy and improved fitness function values, these advantages are offset by longer computation time and possibly a more complex feature set.

4.3.3. Effectiveness of both CTM and ASF

Table 9 shows that IBMRFO outperforms conventional MRFO on several key performance indicators in 4 datasets: average fitness function value, average precision, average number of selected features, and average F1 weighted score. However, IBMRFO shows a slight increase in the average execution time of the algorithm. This trend indicates that while IBMRFO excels in several critical performance metrics, it comes with a slight increase in computational cost. Unlike S-MRFO, IBMRFO shows superior performance in terms of average fitness function value, average precision, average number of selected features on four datasets, and outperforms average F1 weighted score on 4 datasets.

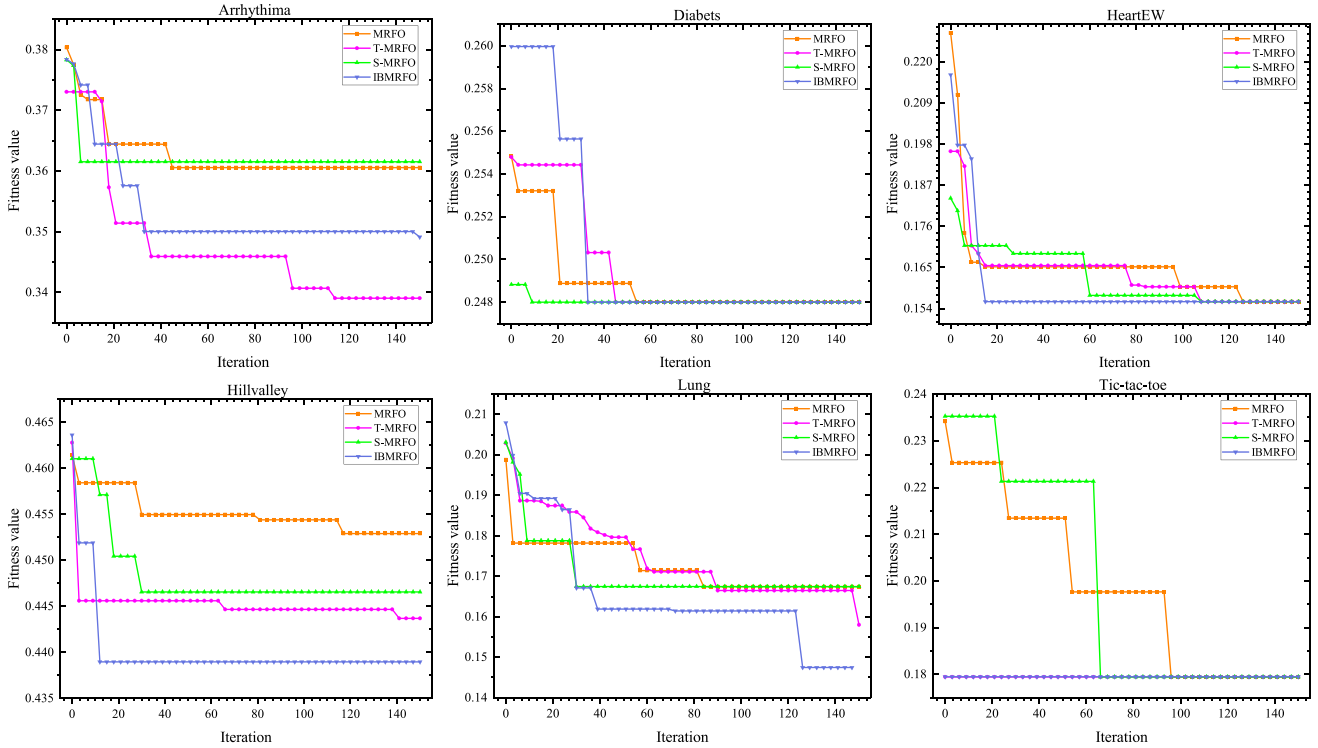
Compared to T-MRFO according to Table 9, IBMRFO achieves better results in terms of average fitness function value, average precision and average F1-weighted score in 3 datasets and outperforms average number of selected features in 2 datasets. However, similar to its performance over S-MRFO, IBMRFO is slightly less efficient in terms of average execution time.

To summarize, the IBMRFO integrates the CTM mechanism and ASF factor into the conventional MRFO algorithm, has significantly enhanced its performance for feature selection tasks. This advancement is evident in various metrics, including improved fitness function value, accuracy, number of selected features, and F1 weighted score.

Table 9

Results obtained by different improved factors.

		Arrhythmia	Diabets	HeartEW	Hillvalley	Lung	Tic-tac-toe
MRFO	fitness $\pm$ std	0.3612 $\pm$ 0.0028	0.2480 $\pm$ 0	0.1564 $\pm$ 0.0013	0.4525 $\pm$ 0.0023	0.1651 $\pm$ 0.0074	0.1795 $\pm$ 0
	accuracy $\pm$ std	0.6407 $\pm$ 0.0027	0.7545 $\pm$ 0	0.8453 $\pm$ 0.0009	0.5475 $\pm$ 0.0022	0.8383 $\pm$ 0.0077	0.8288 $\pm$ 0
	attributes $\pm$ std	152.9 $\pm$ 16.6	4.0 $\pm$ 0.0	4.1 $\pm$ 0.7	45.3 $\pm$ 7.5	164.4 $\pm$ 32.1	9.0 $\pm$ 0.0
	$F1_w$ $\pm$ std	0.549 $\pm$ 0.006	0.751 $\pm$ 0.000	0.846 $\pm$ 0.001	0.545 $\pm$ 0.002	0.821 $\pm$ 0.010	0.816 $\pm$ 0.000
	time $\pm$ std	399.5 $\pm$ 34.9	290.9 $\pm$ 18.6	145.2 $\pm$ 10.0	454.3 $\pm$ 10.5	180.0 $\pm$ 8.3	371.9 $\pm$ 23.3
T-MRFO	fitness $\pm$ std	0.3461 $\pm$ 0.0076	0.2480 $\pm$ 0	0.1560 $\pm$ 0.0008	0.4415 $\pm$ 0.0053	0.1671 $\pm$ 0.0002	0.1795 $\pm$ 0
	accuracy $\pm$ std	0.6537 $\pm$ 0.0073	0.7545 $\pm$ 0	0.8455 $\pm$ 0.0006	0.5563 $\pm$ 0.0051	0.8357 $\pm$ 0.0007	0.8288 $\pm$ 0
	attributes $\pm$ std	91.3 $\pm$ 23.4	4.0 $\pm$ 0.0	4.0 $\pm$ 0.4	21.8 $\pm$ 8.5	125.3 $\pm$ 25.8	9.0 $\pm$ 0.0
	$F1_w$ $\pm$ std	0.571 $\pm$ 0.011	0.751 $\pm$ 0.000	0.846 $\pm$ 0.001	0.554 $\pm$ 0.005	0.833 $\pm$ 0.001	0.816 $\pm$ 0.000
	time $\pm$ std	391.4 $\pm$ 21.1	295.5 $\pm$ 37.8	154.2 $\pm$ 16.1	322.8 $\pm$ 2.3	165.5 $\pm$ 32.0	367.3 $\pm$ 26.0
S-MRFO	fitness $\pm$ std	0.3601 $\pm$ 0.0043	0.2480 $\pm$ 0	0.1565 $\pm$ 0.0014	0.4499 $\pm$ 0.0030	0.1648 $\pm$ 0.0070	0.1795 $\pm$ 0
	accuracy $\pm$ std	0.6415 $\pm$ 0.0040	0.7545 $\pm$ 0	0.8452 $\pm$ 0.0010	0.5501 $\pm$ 0.0031	0.8389 $\pm$ 0.0071	0.8288 $\pm$ 0
	attributes $\pm$ std	144.6 $\pm$ 19.0	4.0 $\pm$ 0.0	4.2 $\pm$ 0.8	45.0 $\pm$ 7.1	172.9 $\pm$ 11.2	9.0 $\pm$ 0.0
	$F1_w$ $\pm$ std	0.551 $\pm$ 0.008	0.751 $\pm$ 0.000	0.846 $\pm$ 0.002	0.548 $\pm$ 0.003	0.822 $\pm$ 0.009	0.816 $\pm$ 0.000
	time $\pm$ std	537.2 $\pm$ 31.3	418.7 $\pm$ 27.4	229.6 $\pm$ 23.7	490.6 $\pm$ 52.9	249.3 $\pm$ 6.1	563.5 $\pm$ 35.7
IBMRFO	fitness $\pm$ std	0.3488 $\pm$ 0.0077	0.2480 $\pm$ 0	0.1560 $\pm$ 0.0008	0.4412 $\pm$ 0.0047	0.1587 $\pm$ 0.0083	0.1795 $\pm$ 0
	accuracy $\pm$ std	0.6508 $\pm$ 0.0072	0.7545 $\pm$ 0	0.8456 $\pm$ 0.0007	0.5566 $\pm$ 0.0044	0.8435 $\pm$ 0.0081	0.8288 $\pm$ 0
	attributes $\pm$ std	85.9 $\pm$ 32.3	4.0 $\pm$ 0.0	4.0 $\pm$ 0.2	23.1 $\pm$ 9.5	123.0 $\pm$ 31.1	9.0 $\pm$ 0.0
	$F1_w$ $\pm$ std	0.567 $\pm$ 0.012	0.751 $\pm$ 0.000	0.846 $\pm$ 0.001	0.555 $\pm$ 0.005	0.827 $\pm$ 0.010	0.816 $\pm$ 0.000
	time $\pm$ std	597.9 $\pm$ 14.9	433.2 $\pm$ 37.1	224.1 $\pm$ 20.6	669.4 $\pm$ 19.0	255.2 $\pm$ 6.0	566.8 $\pm$ 45.5

**Fig. 12.** Convergence rate comparisons between conventional MRFO, T-MRFO, S-MRFO and IBMRFO on different datasets.

While the CTM mechanism contributes to reduced model complexity and overfitting, particularly in smaller datasets, it may occasionally encounter local optima. The ASF factor complements this by improving accuracy and fitness function values, though it increases computational time and feature set complexity. Together, these factors in IBMRFO synergistically balance each other's limitations, offering an effective, balanced approach for feature selection.

5. Conclusion and future works

In this study, we aimed to improve the classification performance of machine learning through optimized feature selection. We proposed

IBMRFO algorithm, an extended version of MRFO that integrates the CTM mechanism, the ASF factor and the S-shaped binary mechanism to improve both exploitation and exploration capabilities while preserving population diversity. This approach effectively balances the two goals of maximizing classification accuracy and minimizing feature count.

Our comprehensive testing of IBMRFO on 30 scientific datasets demonstrated its superiority over other algorithms, achieving better performance in fitness, classification accuracy, F1 weighted scores, and feature subset size, all without additional parameter optimization. These results confirm the effectiveness of our improvements to the MRFO algorithm in feature selection. However, the longer convergence

time indicates an area of potential improvement, perhaps through algorithm refinements or parallel processing techniques. In the future, the IBMRFO will be applied to more complex datasets or specific domains could further strengthen its versatility.

CRedit authorship contribution statement

Kunpeng Zhang: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Yanheng Liu:** Supervision, Funding acquisition. **Xue Wang:** Visualization, Investigation. **Fang Mei:** Formal analysis, Funding acquisition. **Hui Kang:** Resources, Funding acquisition. **Geng Sun:** Writing – review & editing, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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